

MONOLYTHIUM

Whitepaper

V5.1 · JUNE 2026

Settlement Layer for the Autonomous Economy

NETWORK

Monolythium · LYTH

STEWARDED BY

Mono Labs R&D LLC · Monolythium Foundation

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v6 reconciliation — updated 2026-07-16. This is the v5.1 text (June 2026) with a set of factual corrections applied so that it matches the running Monolithium v2 chain (LythiumDAG-BFT) as of the **v0.4.0 re-genesis of 2026-07-07**. The affected sections have been rewritten in place; a full v6 edition will re-scope the surrounding narrative. What changed relative to v5.1, and why:

1. **Cross-chain interop is now an external-provider integration, not an in-tree bridge.** The entire in-tree bridge stack — the on-chain bridge proof verifier, the route-policy, fee, and insurance scaffolding, and the associated build feature — has been **removed** from the protocol. The chain no longer verifies bridge proofs on-chain. Interoperability with external chains is instead delivered by integrating an **external interop provider** (evaluation in progress; described vendor-neutrally throughout). This moves the highest-risk, audit-heavy surface out of consensus and lets interop evolve without a re-genesis while the core stays post-quantum and lean. §20 is rewritten accordingly and the passing “zero-knowledge / light-client bridge” references elsewhere are updated.
2. **The application-layer Groth16-BN254 / SP1 zero-knowledge verifier is disabled at genesis.** The intended direction is a post-quantum FRI/STARK verifier that ships **gated off** until it is ready; until then no application-layer ZK verifier is reachable on the value path. As stated in §12.4, consensus finality is pure ML-DSA-65 and never depended on it.
3. **The encrypted mempool (“LythiumSeal”, §12.5) has been removed.** v2 runs a **plaintext** mempool; ordering fairness is handled at the DAG-consensus layer. The threshold-decryption sealing design was dropped — it relied on the same shared/threshold-key machinery that per-operator DVT multisig replaced.
4. **DVT uses a per-operator ML-DSA-65 multisig, not shared or threshold keys.** Any earlier FROST/DKG/threshold-BLS shared-key construction is gone: each operator signs with its own independent ML-DSA-65 key and a cluster certificate is a 7-of-10 bitmap of those raw signatures, with no shared key and no key-setup ceremony.
5. **Cross-cluster finality quorum is count-based, not stake-weighted.** An anchor is final once a $2f+1$ **count-based** quorum of clusters (each with equal weight) has contributed a valid 7-of-10 cluster vote. Stake sets only the top-100 admission rank (§16, §17), never vote weight.
6. **Confidential (amount-hiding) value transfer has been removed from the protocol and is not live.** Stealth-address recipient privacy and per-account / per-asset privacy policy are live; confidential amounts (the public→private crossing and the shielded spend/exit value path) are **not implemented**. The

earlier Ristretto/Pedersen + Bulletproofs construction was **deleted** rather than left disabled: an audit found its commitment path could mint unbacked value, and a disabled precompile is a weak guarantee when a signed configuration entry could re-arm it. Confidential amounts are now described as a future post-quantum capability rather than a present-tense or gated feature (§5, §12, §19).

7. **The on-chain prover / GPU proof market and service-tier oracle-feed payments (§16.3, §17.3, §17.4) are gated off at genesis** and are not reachable in the public v0.4.0 protocol. They are described as roadmap capabilities.
8. **Wallet recovery uses a standard 24-word BIP-39 mnemonic** re-derived for ML-DSA-65 as `mldsa_seed = SHAKE256("monolythium.mldsa65.v1" || bip39_pbkdf2_seed(mnemonic, ""))[0:32]`. The earlier “PQM-1” mnemonic format was dropped; the §13.3 PQM-1 domain string and seed-layout table are **superseded** (see the banner in §13.3) and must not be used to derive addresses.
9. **The registered v0.4.0 development-network topology is a 4×10 DVT fleet** (four clusters of ten operator identities, 7-of-10 each) plus two relay identities — 42 registered nodes — not a single cluster. That identity and topology record is not proof that the network is currently advancing; see the dated degraded-status observation below. If all four clusters are participating, the cross-cluster quorum is $2f+1$ with **f = 1**.
10. **The SLH-DSA hash-based emergency-backup key and the in-protocol emergency-key registry are removed; there is no on-chain emergency algorithm rotation at launch.** The chain is **ML-DSA-65-only**. The advertised break-glass — a pre-registered SLH-DSA (FIPS 205, hash-based) backup key, the on-chain emergency-key registry, and the algorithm-rotation event that would activate it — was found in audit to be inoperative and has been **removed** from the protocol. Post-quantum **crypto agility** (a second-family backup primitive plus an in-protocol rotation path) is now described as a **future capability to be introduced in a subsequent genesis**, not a present-tense feature; §7 (“the quantum hedge”), §12.1, §22, and §23.1 are corrected accordingly. The emergency **freeze** circuit breaker (a global, time-bounded pause, §23.2) is a separate mechanism and is unaffected.
11. **The classical signature code has been removed from the implementation, not merely refused at admission.** §12.1 already stated that ML-DSA-65 is the only signature primitive accepted at transaction admission, and that has always been enforced. But the implementation still carried the classical primitives — secp256k1, Ed25519, WebAuthn P-256 — behind a build flag, along with a classical/post-quantum **hybrid** construction and a wallet-side key-import path for them. None of it could produce a transaction the chain would accept, and §7 argues at length that a hybrid mode should not exist; carrying the code anyway was the gap between the paper and the tree. It is

now deleted: the primitives, the crate that hosted them, the hybrid construction, the build flag, and the import paths. Their wire tags are **permanently retired** — a transaction presenting one resolves to no algorithm and is rejected before verification — and the tags are never reused, so every remaining algorithm’s wire value is unchanged. The claims in §7 and §12.1 (“no classical fallback, no hybrid mode, no ECDSA acceptance path”) are therefore now true of the code and not only of the admission gate, and §12.5’s statement about the build-enforced lint is corrected to match: there is no longer any allow-listed signature or KEM exception. One classical dependency remains and is disclosed in §12.3 — the peer-to-peer transport’s node-identity keys, which authenticate network connections and never consensus or state. The honest claim is **no classical primitive in the protocol path**, not “no classical code anywhere in the build.”

12. **Two long-standing descriptions of the cryptography were wrong and are corrected: the peer-to-peer handshake, and the argument against hybrid signatures.** §12.3 described the operator-to-operator handshake as “the Noise XX handshake” with ML-KEM-768 replacing X25519. It is not Noise XX and it is not a stock Noise pattern at all: libp2p’s Noise implementation hardcodes X25519 with no plug-point for a different key exchange, so the chain ships a **self-contained KEM-Noise IK-style handshake of its own composition**, which has **not yet had an independent cryptographic review** — that review is a tracked release gate before mainnet. §12.3 now says so, names the actual construction, and discloses the two classical artefacts that remain at the transport layer (a linked-but-never-invoked X25519, and the ed25519 key libp2p requires for `PeerId`). Separately, §7’s first argument against hybrid signatures was unsound: it defined hybrid as consensus “accepting either” signature but then described a forgery that only makes sense against a scheme requiring **both** — and that forgery does not work against that scheme, because a stale post-quantum signature does not verify over a new message. The conclusion (pure post-quantum, no hybrid) is unchanged; the reasoning now treats the two hybrid shapes separately and states honestly which risk a classical half would and would not hedge.
13. **The eight-primitives agent-commerce suite is a target architecture, not an activated public- testnet capability.** No current public release provides protocol-enforced service listings, availability, escrow, arbitration, bilateral reputation, agent spending policy, economic runbooks, or autonomous signing. Section 18 now states this boundary explicitly. Its detailed state machines and method names are design requirements, not a current ABI or integration guide.
14. **Stele’s product architecture is standalone web, not an embedded desktop-wallet marketplace.** As of 2026-07-16, the public preview is live at stele.monolythium.com with zero published services. Browser Wallet v0.4.5 is a prerelease; Stele’s hosted services never receive wallet secrets. The

public web authenticates users through Browser Wallet. That authentication proves only current control of the selected wallet address; it does not prove a human or legal identity, durable ownership, credentials, or authority. Its wallet-authenticated Provider Studio at stele.monolythium.com/studio can create, edit, preview, and delete private wallet-owned provider-listing drafts; these durable provider-listing drafts are not published, discoverable, or transactable, and provider publication remains off. Booking-approval drafts are separate: the web can inspect an existing valid non-economic booking-approval draft, but it does not create booking-approval drafts. Inspection is read-only and non-economic: the public web exposes no booking, payment, settlement, or other economic controls. Hosted Stele MCP is keyless and OAuth-protected. Its authenticated `tools/list` response is the authoritative inventory for that session. Its baseline tools provide public catalog search and bounded, non-economic booking-draft preparation; hosted booking-draft preparation is unavailable without a published listing. The gated `stele_create_provider_listing_draft` tool exists only when that exact name appears in authenticated `tools/list`. When listed, each successful `stele_create_provider_listing_draft` operation creates exactly one new private wallet-owned unpublished provider-listing draft. An idempotent replay returns the same draft without creating another. The tool cannot list, read, update, delete, or publish an existing draft. Hosted MCP does not sign, submit, broadcast, or settle transactions. A separately installed local Stele MCP exposes exactly three read/status tools and no transaction tool. Economic writes, transaction signing, and mainnet remain off. The current product line retires embedded Stele, but legacy or unreconciled desktop builds may retain a gated historical surface until their separately reviewed removal and migration are released. E2EE rooms, settlement, disputes, and reviews remain separately gated. See §18.10 for the product boundary.

15. **Consensus leader selection is a deterministic round-robin schedule.** The leader-selection design is the Starfish pacemaker's round-robin: the wave leader is `wave number mod active-cluster-count`, computed from public DAG state with no seed and no separate leader-selection signature. §11 previously carried a second, contradicting description — a “post-quantum leader-seed beacon” derived from a BLAKE3 hash of the quorum certificate — as a superseded design that the text never removed; §11 and §22.3 are corrected to the round-robin design. Its honest anti-grinding property is that there is no aggregate seed for an adversary to bias — leader-grinding is foreclosed by construction — but leadership is **public and predictable**, so the earlier “unforgeable leader selection” framing is dropped, and transaction ordering within a leader's proposal is named as a separate consideration the schedule does not foreclose.

Dated network-status note — 2026-07-16. The registered v0.4.0 public development-network identity remained reachable, but its public RPC reported height 74,907 with latest timestamp 2026-07-08T22:39:05Z and did not advance during repeated checks. It must therefore be treated as **degraded/stale**, not as liveness evidence or a production settlement network. Any undated present-tense wording retained in historical discussion does not override this observation.

“Sovereignty is not given; it is verified by the silicon and the math.”

Abstract

Monolythium is a Layer 1 blockchain designed as a settlement layer for the autonomous economy — a future in which humans, companies, AI agents, machines, and software services transact directly on open rails.

The chain is built around six positions:

- post-quantum accounts by default, with no classical signature acceptance path;
- Rust-first smart contracts compiled to a deterministic RISC-V execution target;
- native modules for tokens, NFTs, and markets, with payments and agent commerce as separately activated target modules;
- a structurally bifurcated denomination that separates monetary privacy from commerce so the chain cannot be used as a fungible-anonymous-payment rail;
- a cluster marketplace that turns validator operation into a public, competitive market with distributed validator technology;
- no on-chain governance and no perpetual futures or margin, so the surface that can be captured, gamed, or weaponised is smaller.

Monolythium is not designed to be another EVM-compatible chain, and it is not designed to compete with the open agent-payment standards now shipping at scale (x402, AP2, ACP, MCP, Visa Intelligent Commerce, Mastercard Agent Pay). Its target is to **settle underneath** them: a chain-anchored trust, policy, escrow, identity, and reputation layer that compatible rails could compose against after activation.

The public position is:

Monolythium is not EVM-compatible at execution. Its target architecture connects at the liquidity edge and aims to become composable underneath compatible agent-payment standards after each required integration is released.

The target architecture allows assets to arrive through independently released external-provider or issuer integrations; the chain no longer ships an in-tree cross-chain bridge. Agent-payment standards and the eight agent-commerce primitives describe the intended composition layer. They are not activated settlement paths on the current public development network.

This document has three parts.

Part 1 — Why Monolythium explains the design philosophy: what the chain is for, the first commercial wedge, how it composes with existing payment standards, what it refuses, and the structural choices that follow from those refusals. It is written to be read end-to-end.

Part 2 — How Monolythium Works is the technical reference: consensus, cryptography, identity, execution, native modules, tokenomics, cluster operations, agent-commerce primitives, privacy, interop, hardware, threat model, and recovery. It is written to be read in sections.

Part 3 — Adoption and Outlook covers developer experience, user experience, honest limitations, and what success looks like.

Contents

Part 1 — Why Monolythium

1. The Thesis: Settlement Layer for the Autonomous Economy
2. The First Wedge
3. Target Composition with Agent-Payment Standards
4. What Monolythium Refuses
5. Bifurcated Denomination: Privacy Without Contamination
6. Cluster Marketplace: Validator Operations as a Public Market
7. Post-Quantum from Genesis
8. Forking as the Exit Path
9. Why Rust and RISC-V
10. Market Positioning

Part 2 — How Monolythium Works

1. Consensus: Starfish-C
2. Cryptography
3. Identity: Addresses, Mnemonics, and Names
4. Execution: Rust on RISC-V
5. Native Modules and MRC Standards
6. Tokenomics
7. Cluster Operations: DVT, Slashing, Service Tiers
8. Target Agent-Commerce Architecture
9. Privacy Cordon
10. Interop and the Liquidity Edge
11. Hardware Sovereignty
12. Threat Model
13. Recovery and Emergency Posture

Part 3 — Adoption and Outlook

1. Developer Experience
2. User Experience
3. Honest Limitations
4. What Success Looks Like
5. Closing

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PART 1 — WHY MONOLYTHIUM

1. The Thesis: Settlement Layer for the Autonomous Economy

Most production blockchains today are honestly named **trading chains**. They optimize for the round-trip latency of a swap, the throughput of a perpetuals exchange, or the fee surface of a high-frequency strategy. Trading is fine; trading is not what the next ten years of digital economic activity will primarily look like.

The next ten years are **agentic**. An AI agent that pays for inference, hires a human contractor, buys data, settles an invoice, runs a marketing workflow, escrows funds against a legal deliverable, subscribes to compute, or pays another agent for a service is not science fiction. Each of those actions is technically possible today; what is missing is open, neutral, enforceable rails on which to settle them.

Today, an agent that wants to spend money does so through an account it shares with its user inside a closed AI platform, or through a payment-processor account leased by the user, or through a custodial wallet rented from a single exchange. Each of those rails is **owned**. The agent is rentable, the identity is rentable, and the reputation accrued through years of work is rentable. If the platform changes its terms, the agent's economic life resets.

Monolythium exists to provide a different option. The chain is built around the assumption that the most interesting category of economic activity in the next decade will be **delegated** — a human or an organization authorizes a software entity to act on their behalf, with bounded authority, against open infrastructure, across providers and jurisdictions. The substrate for that activity needs to be:

- programmable, so policy is enforceable in code;
- auditable, so principals can verify what their agents did;
- permissionless, so an agent cannot be deplatformed by a single counterparty;
- interop-connected, so it connects to existing value through an external interop provider;
- post-quantum, so identity survives multi-decade cryptographic horizons;
- and structurally hostile to use cases that depend on combining anonymous payment with anonymous service discovery.

The chain is useful even if the agentic category grows slowly. Native tokens, NFTs, spot markets, payments, and application-specific contracts are valuable in their own right. The strategic focus is to remain genuinely useful in both the success case (agent commerce becomes a major category) and the failure case (it grows slower than expected), and not to take design decisions that only pay off if a single optimistic scenario plays out.

The chain is the same chain it would be if no agent ever showed up. The user is different. The agent is the audience the chain quietly fits best, and quietly was designed for from the start.

Target behavior for an agent

The following paragraphs describe an intended end state. Agent sub-accounts, protocol spending policy, economic runbooks, escrow, arbitration, and settlement are not activated on the current public development network.

In the target model, an agent has a chain-native sub-account address under a human or organizational principal. The agent itself is not a legal entity. The principal would issue a programmable spending policy—budget caps, allow-listed counterparties, time windows, and per-call limits—that a future activated protocol version enforces in consensus state.

The target model uses canonical **runbooks**—typed, versioned templates for supported operations—instead of arbitrary raw RPC. Their vocabulary is not a current executable API.

The target model derives role-separated, settlement-bound reputation from finalized history so that a future agent can retain its record across model providers. No current public reputation surface should be inferred from this design.

If activated, the same agreement spine could serve a human hiring an agent, an agent hiring a human, agents hiring each other, or two humans: offer, counter-offer, escrow, delivery, release, and dispute. The arbitration modes remain design requirements pending a versioned protocol contract.

An illustrative future example

Imagine a future small business giving an assistant a dedicated sub-account with a bounded policy. The assistant searches a finalized service registry, prepares exact terms for a lawyer, and negotiates a timeline. Funds move only after the activated protocol, SDK, wallet, and user policy all accept the same typed transaction. Delivery stays private; release or dispute produces a finalized receipt.

The lawyer is human and the hiring decision is delegated, but this remains a target scenario—not evidence of a current offer, escrow, settlement time, reputation update, or fee schedule.

The 30% cut also paid for real services on existing platforms — discovery, credential vetting, dispute resolution, consumer protection, and demand aggregation. The target architecture would replace or complement each function rather than pretending the function disappears:

- **Discovery** would use a permissionless registry plus competing indexers; any anti-spam bond remains release-gated.

- **Credential vetting** would use versioned issuer and attestation primitives while markets decide which issuers they trust.
- **Dispute resolution** would use released escrow and arbitration contracts appropriate to the value at stake.
- **Consumer protection** would combine principal-owned approvals, activated policy enforcement, denomination constraints, and accountable arbitration.
- **Reputation** accrues to the provider's chain address as portable, multi-dimensional signal that survives any single venue.

The protocol design does not solve **demand aggregation**—getting buyers to a future registry in the first place. That remains an application-layer problem. Steele may aggregate public catalog demand, but its current catalog is not proof of a chain-native listing.

That story has many variants — replace the lawyer with another AI agent, replace the business with a household, replace the legal task with a compute job or a marketing campaign or a kitchen renovation — and the chain primitives are the same. The breadth is the point. **Universal services on a single primitive set, not domain-specific marketplaces stacked on top of a generic chain.**

The agent-wallet moment

The simplest way to understand the category is this:

The moment an AI assistant can say, “Send me 50 USDC to complete this task,” the agent-commerce era has started.

In the target architecture, that would not give the agent unlimited authority. A user could give an agent a dedicated account, budget, and policy; the agent could request funds or prepare a bounded action, while a released wallet and activated protocol enforced the user's rules. Current Steele integrations do not give an assistant a wallet or transaction tool: hosted MCP is keyless, the isolated local MCP is read/status-only, booking-approval drafts are non-economic review records, and booking, payment, and settlement controls are unavailable.

This is different from AI-assisted checkout. A checkout protocol lets an assistant help a user buy from a single merchant. Monolithium's surface is broader: an assistant can have an economic account and operate under user-defined rules across many services, across many providers, without depending on any individual marketplace.

What the chain does not over-claim

The bet that an open, neutral settlement layer wins as the trust layer behind the agentic economy is not a sure thing. Existing payment networks, large model providers, and cloud platforms are shipping their own agent-payment infrastructure today. The chain's

bet is not that those rails disappear. The bet is that **the rails will need a neutral, chain-anchored substrate beneath them** for the parts they cannot supply themselves — enforceable principal-level policy, escrow with dispute resolution, portable cross-platform reputation, and post-quantum-grade long-lived identity.

The chain is designed to be useful in either outcome. Spot markets, post-quantum settlement, bifurcated privacy, and external-provider interop remain valuable on their own and would not need rebuilding if the agentic category grew slowly.

2. The First Wedge

A settlement layer for the autonomous economy is the long horizon. The first commercial wedge — where the chain enters real workflows — is narrower.

The planned wedge is agent escrow, spending policy, and reputation as the trust layer behind paid APIs and services.

The market is already converging on standardized payment handshakes for AI agents. Coinbase shipped x402 in 2025; Google's AP2 launched the same year; Stripe and OpenAI released the Agentic Commerce Protocol; AWS Bedrock AgentCore added native stablecoin payment rails through Coinbase and Stripe; Visa and Mastercard both announced agent-payment programs. These standards solve the **payment handshake** — how an agent discovers a paid endpoint, presents payment, and receives the response.

They do not solve four problems that are unavoidable once agents operate against more than trivial workflows:

- **Enforceable principal-level spending policy** — caps, allow-lists, time-of-day rules, expiry, revocation — that the agent itself cannot bypass. A wrapper around a card or a custodial wallet enforces policy at the wrapper; if the agent is granted the wrapper's credentials it inherits its full authority. The unactivated vNext design would enforce policy at admission, in consensus state, on a sub-account the principal controls and the agent cannot rewrite.
- **Escrow with counter-offer flow and pluggable arbiters** for non-trivial work — the deliverable-against-payment use case that defines real services. A paid-API handshake does not negotiate timelines, attach penalty clauses, or arbitrate disputes; an escrow primitive does.
- **Portable cross-platform reputation tied to the agent identity** — speed, quality, communication, accuracy — that survives the agent switching model providers, runtimes, or wallets. A closed platform's reputation cannot follow the agent off the platform.
- **Chain-anchored consent records** the principal can revoke instantly, leaving in-flight commitments grandfathered. A consent stored in a vendor's database can be overwritten by the vendor; a consent stored on-chain has the principal's signature on it.

The target architecture assigns those four primitives to versioned native or indexed surfaces. They are planned composition points, not current public-testnet deliverables. A future x402, AP2, ACP, or MCP integration may use them only after its exact protocol, asset, wallet, privacy, and connector contracts are released.

If released, the wedge is narrow enough to measure. Candidate adoption signals include:

- the count of agent sub-accounts created against Monolythium spending policies;

- the volume of escrow flowing through Monolythium for ACP/AP2/x402-mediated transactions;
- the depth of the discovery registry and the number of legitimate providers it hosts;
- the survival of agent identity and reputation across changes in model provider, wallet, or platform.

If those metrics grow, the broader settlement-layer thesis follows. If they do not, the chain still has its native markets, post-quantum identity, bifurcated denomination, and interop surface — none of which depend on the agentic wedge succeeding.

3. Target Composition with Agent-Payment Standards

Monolythium is designed to compose **underneath** the open standards now shipping for AI-agent payments. The table describes intended integration roles after each relevant protocol, SDK, asset, wallet, and privacy contract is released. It is not a list of current integrations or activated settlement paths.

LAYER	STANDARD	POSSIBLE MONOLYTHIUM ROLE AFTER ACTIVATION
HTTP payment handshake	x402 (Coinbase, x402 Foundation)	Could bind x402-initiated settlement to a versioned receipt, escrow, and reputation model
Agent intent mandate	AP2 (Google, open)	Could anchor a mandate to a principal-controlled consent and spending policy
Checkout flow	ACP (Stripe, OpenAI)	Could bind checkout output to settlement, counter-offer, and dispute state
Tool / connector boundary	MCP (Model Context Protocol)	Could expose released discovery or approval workflows; transaction tools require a separate local signing design and explicit user authority
Card-issuer agent flows	Visa Intelligent Commerce, Mastercard Agent Pay	Could add portable identity, receipts, and reputation behind issuer-controlled payment flows

The intended composition is one-directional: external rails would initiate payment, transmit intent, or complete checkout, while a future activated Monolythium surface could add policy, escrow, dispute, reputation, and settlement. No current rail should treat this design as an integration claim.

Illustrative future composition: x402 + Monolythium

The following is a design scenario, not a runnable flow. Suppose an assistant requests a paid API, the endpoint returns `HTTP 402`, and a future wallet supports a principal-owned Monolythium agent account with an activated spending policy:

1. The wallet would check the released policy schema against category, per-call cap, and aggregate cap.
2. After explicit authority under the released signing contract, the wallet could sign a typed transaction referencing the external payment identifier.
3. An activated protocol version would reject policy violations at admission.
4. A released receiver could wait for observed finality and fetch the versioned receipt.
5. A finalized event could feed the principal's audit trail.

6. If the released escrow and arbitration protocol supported the case, the principal could open a dispute tied to that receipt.

The safety property sought by this scenario is that the wallet, typed action, and protocol carry the discipline rather than free-form model output. It does not assert a current x402 receiver, `pay_vendor` ABI, stablecoin route, settlement-time SLA, or dispute integration.

Illustrative future composition: AP2 + Monolithium

A future AP2-compatible flow could bind a principal-signed mandate to a released Monolithium consent schema. This is also a target scenario, not a current integration:

1. A future wallet could register a mandate with a narrow scope and expiry under a released consent schema.
2. The assistant could prepare merchant terms without receiving signing authority.
3. The wallet could compare the proposed action with the mandate before the user authorizes a transaction.
4. An activated protocol could bind the resulting receipt to the consent version.
5. A released revocation rule could reject later uses while preserving already-finalized commitments under an explicitly documented policy.

The target property is principal-owned revocation with a verifiable authorization history. The exact grandfathering rule, merchant contract, and AP2 mapping remain release-gated design work.

Why the composition stance

Two reasons.

The major rails will route value. Every credible projection of agent-commerce volume assumes the existing payment networks (cards, ACH, stablecoins over x402) move most of the flow. A chain that tries to displace them at the handshake layer fights a market with deep distribution, mature compliance, and large customer balances. A chain that settles **underneath** them provides what the handshakes cannot supply and benefits from the volume regardless of which handshake wins.

The trust gap is real. A payment handshake solves payment. It does not solve the agent's authority, the principal's revocation, the deliverable's enforcement, or the reputation's portability. The target architecture is designed to address those gaps after activation; they get harder, not easier, as agent volume grows.

The chain is not intended as an alternative to x402, AP2, ACP, or MCP. The thesis is that a future released Monolithium layer can supply settlement, policy, escrow, identity, and reputation where those rails expose a compatible integration boundary; that thesis is not proof that any named rail has adopted it.



4. What Monolythium Refuses

A chain's identity is shaped at least as much by what it refuses to do as by what it does. Five refusals define Monolythium.

4.1 No on-chain governance

There is no governance token, no on-chain proposal contract, no protocol-level voting mechanism, no on-chain treasury voting, and no signaling extrinsic that pretends to be governance.

The reason is attack surface and clarity. On-chain governance, as deployed across a decade of L1 experiments, has consistently produced four failure modes:

- **Capture.** Large holders, custodians, and exchanges accumulate vote weight and steer outcomes.
- **Bribery markets.** Voting rights become rentable; outcomes follow whichever vote-buying market is liquid that week.
- **Theatre.** Low participation means a small, motivated minority decides everything, while the rest of the network's name is attached to the decision.
- **Direct control surface.** A working governance contract is a one-step path for an attacker to alter the protocol if they obtain a quorum of votes — by exploit, by accumulation, or by social engineering.

The chain rejects all four by not having the mechanism. There is nothing to capture, nothing to bribe over, nothing to vote against an empty hall about, and nothing to steer if you compromise a quorum.

The replacement model is honest about its centralisation in one place and aggressively decentralised in another:

- Protocol direction is set by accountable maintainers, in public, with explicit rationale.
- Community input is gathered through public channels and weighed.
- Operators choose whether to run new versions of the node software.
- The legitimate path of dissent is **forking**, covered in §8.

This is more honest than pretending a 4%-turnout vote is a popular mandate. It draws a clear line between **direction** (decided by maintainers and operators) and **consent** (expressed by running software, holding the asset, or forking off).

4.2 No perpetual futures or margin

The chain has spot markets. The chain does not have, and is not building, perpetual futures, margin trading, or any on-chain leverage venue. The protocol does not host a perpetuals exchange and does not mint a tradeable perp asset class.

The reason is product identity. Once a chain becomes a perpetuals venue, the design pressure on every other surface bends towards the perpetuals product — oracle dependencies, liquidation engines, funding-rate disputes, MEV around liquidations, leverage-driven liquidity, and a regulator audience that classifies the chain as a derivatives exchange whether it likes it or not.

Activated spot markets, payments, NFTs, cross-chain routes, and conservative DeFi primitives can form a deep ecosystem on their own. Agent-service settlement, bounded budgets, escrow, and delegated policy remain unactivated vNext targets; they must not be inferred from those existing primitives.

This refusal also reduces the audit surface dramatically. A perpetuals venue is one of the most complex artifacts in DeFi; the chain removes that complexity entirely from the consensus-critical surface.

4.3 No EVM execution

Monolythium is **not EVM-compatible at execution**. Existing EVM bytecode does not run on the chain. Solidity is not the default developer model. The token, NFT, and account-abstraction conventions from the Ethereum ecosystem are not the chain's protocol standards.

This is a deliberate position. The chain is **EVM-connected at the liquidity edge** — value moves in and out through an external interop provider and issuer-supported integrations, rather than through an in-tree cross-chain bridge — but value, once it arrives, settles through Mono-native standards on a Rust/RISC-V execution layer.

The reasoning is in §9. The short version: trying to be a faster Ethereum has been tried and produces well-funded chains that nevertheless live inside Ethereum's frame. The chain optimizes for the audience that wants Rust contracts, AI-assisted developer ergonomics, post-quantum accounts, native modules for hot paths, and a smaller and cleaner protocol surface.

4.4 No fungibility between public and private money

The target design has two denominations of LYTH: **public** and **private**, with no reverse private-to-public crossing. The confidential value implementation that would create and move amount-hidden private LYTH was removed after audit, so users cannot make this crossing today.

Any future replacement is constrained to private transfer and burn and must remain unable to enter contracts, markets, interop, delegation, or a service-discovery module. Those are design constraints, not current transaction types.

This is the most defensible privacy posture available to a general-purpose L1. It is detailed in §5 and the cordon implementation in §19.

Implementation status. The confidential value path that implemented the public→private crossing, private transfer, and private burn was **removed**, not merely gated. Stealth-address recipient privacy and per-account / per-asset privacy-policy code remain in v0.4.0; the stale public RPC observed on 2026-07-16 is not liveness evidence. Any confidential-value replacement requires a new post-quantum construction and separate activation (§19).

4.5 No bundled AI model

The chain does not bundle a particular AI model with the protocol. The unactivated vNext design makes agent identity provider-independent and requires typed, signed runbooks rather than “AI prompts”; neither agent sub-accounts nor economic runbooks are current public-network capabilities. The architecture remains model-agnostic so that a later activated surface can support open-source models, proprietary models, narrow domain-specific systems, and non-AI software agents alike.

This refusal matters because it preserves the chain’s neutrality. A protocol that bundled a specific lab’s model would tie its reputation, longevity, and credibility to a single vendor’s roadmap. The agent economy needs the opposite — a substrate the model providers can plug into rather than one that picks the winning model in advance.

5. Bifurcated Denomination: Privacy Without Contamination

Privacy on a public ledger has, over the last several years, been on a steady regulatory collision course. The two historical trajectories are visible.

A chain with fungible shielded and transparent denominations — funds can flow freely between shielded and transparent and back — ends up delisted by major exchanges, because shielded-to-transparent flow contaminates the entire public denomination. An auditor cannot distinguish a deposit that came directly from a transparent address from a deposit that came from a transparent address that was funded yesterday by a shielded address that was funded the day before by a transparent address that was funded by an address on a sanctions list. Fungibility breaks the audit story for everyone.

A chain with a single private-by-default denomination ends up delisted by major exchanges, because there is no public denomination to clear. The audit story is “we cannot tell you anything.” Exchanges respond by removing the asset entirely.

Both trajectories converge on the same outcome: a privacy-supporting L1 ends up delisted, regardless of which mechanism it used. Monolithium’s bifurcated denomination is a third option that has not been deployed at scale by a major L1.

Design-status boundary. The confidential (amount-hidden) value path—public→private crossing, private transfer, and private burn—was **deleted after audit**. This section preserves the intended cordon and economic separation as requirements for any future post-quantum replacement. It does not describe usable private value today.

How bifurcation works

The target protocol enforces non-fungibility at the consensus layer with separate public and private state and a native caller-origin cordon (§19). The cordon forbids a public contract or module from receiving private-denominated value and forbids private state in a public contract context. Stealth-address recipient privacy and privacy-policy code remain; confidential amount transfer does not.

If a future post-quantum value path is activated, a **crossing** must decrease public value and create the same private-denominated value exactly once, with no reverse instruction. The removed construction must not be re-armed; the replacement requires independent conservation proofs and audit evidence.

What this gives a regulator or an exchange

A regulator or exchange treats the **public denomination** as fully auditable. Public activity is transparent, traceable by the analytics tooling that already exists for other transparent chains, and visible to subpoena. A KYC-supervised exchange can custody public LYTH with the same audit posture as it custodies any other transparent asset.

A regulator or exchange treats the **private denomination** as opaque by design. They are correct to do so. The protocol does not pretend otherwise. An exchange may choose to refuse private-denominated deposits entirely; an exchange may choose to require a “proof of crossing” before accepting them (the user demonstrates, via an optional application-layer proof, that the private LYTH crossed from a public address that was KYC’d at the time of crossing); an exchange may choose to accept private deposits at a higher KYC tier. The chain does not constrain the choice. The chain provides the **structural separation** that makes the choice coherent.

A user who values privacy crosses LYTH into the private denomination and uses it for peer-to-peer transfers. They cannot pay an exchange fee with private LYTH; they cannot invest private LYTH in a contract; they cannot hold private LYTH on a centralized venue that does not accept it. They retain monetary privacy for their direct transfers. The privacy is real. The cost is structural — privacy-denominated value cannot participate in the broader chain economy, by design.

Why this is the position

The regulatory soundbite, stated plainly:

Monolythium separates monetary privacy from commerce. The private denomination supports peer-to-peer transfers only — it cannot pay for services, interact with smart contracts, or trade. The public denomination supports full commerce and is fully transparent and analyzable. This bifurcation makes the chain structurally hostile to use cases that depend on combining anonymous payment with anonymous service discovery. Compliance happens at frontends and fiat boundaries; the protocol stays neutral.

It is true, defensible, and not common in production.

Side effect: structurally hostile to illicit commerce

The pattern that history calls “the darknet marketplace” requires the simultaneous availability of an anonymous-payment rail and an anonymous-service-discovery surface. The target bifurcation forbids private-denominated value from any future discovery, escrow,

or dispute module. Public Steele catalog data is transparent and non-economic; it is not an activated on-chain discovery registry. Any future confidential-value and commerce modules must preserve this cordon before activation.

This is not the only defense the chain offers against illicit commerce — frontend curation refuses to surface illegal listings, arbiters refuse to arbitrate illegal disputes, attestation tilts markets toward legitimate providers, and fiat on-ramps apply KYC as they would for any other asset. But the bifurcation is the structural defense, the one that does not depend on application-layer compliance or post-hoc enforcement. It is by construction.

Institutional accumulators

A common question: what happens when an institution wants to accumulate LYTH privately for treasury purposes? Bifurcation answers cleanly. The institution holds public LYTH in a transparent address; if they want to obscure their accumulated balance from competitors, they cross LYTH into private and hold the private balance. The private balance is opaque to external observers but is also locked from any chain-native productive use — it cannot stake, cannot delegate, cannot earn yield from a contract. The institution faces a real trade-off: privacy on treasury holdings or chain-native productive use, but not both. They can split: hold a productive position publicly and a private reserve privately.

6. Cluster Marketplace: Validator Operations as a Public Market

Monolythium does not have validators in the traditional single-operator sense. It has **clusters**: 7-of-10 operator-threshold groups that produce one logical cluster vertex per cluster, per consensus round. A cluster is staffed by ten independent operators, runs continuously, and tolerates up to three operator outages without missing a beat.

The full cluster set at the target scale is **100 clusters × 10 operators = 1,000 operator positions across the network**, with seven active and three standby positions per cluster (1,000 = 700 active operator seats + 300 standby).

The reasoning is direct: a single validator per stake-weighted position is a single point of failure, a single point of slashing, and a single source of geographic, network, and jurisdictional risk. A cluster is structurally less brittle in all three dimensions.

The marketplace

Cluster membership is a public market. Operators publish their **node attestation** (hardware class, network metadata, geographic claim, uptime track record, service-tier capacity), and clusters compose themselves from operators offering complementary skills:

- a bare-metal operator in one region for raw throughput;
- a cloud operator in a second region for geographic and network diversity;
- a home operator in a third region for jurisdictional diversity;
- a GPU-equipped operator for the cluster's prover service tier;
- an archive-capable operator for the cluster's RPC and archival service tier.

Clusters compete on reliability, latency, uptime, geographic diversity, service-tier breadth, hardware quality, and reputation. Stakers see all of this through wallet and explorer surfaces and delegate accordingly. The “best validator” is not picked by social reputation; it is composed in public by people who can read the same data the chain reads.

This is sometimes shorthand as the “**Avengers assembly**” model — a real differentiator. Most chains hide operator selection behind opaque delegation lists in a wallet; Monolythium turns the composition itself into a public market with first-class wallet, explorer, and reputation support, so operator quality is visible to delegators rather than tucked behind a brand.

Why this matters for agent commerce

Agents need reliable infrastructure. A chain designed for autonomous settlement should make infrastructure quality visible and economically meaningful, so that operator quality is a market signal rather than an opaque social fact. Cluster reputation is the visible quality of the substrate the agent runs on. Operators with strong uptime, broad geo-

graphic coverage, and rich service-tier offerings command premium delegation; operators that go offline or run weak hardware lose market position. The agent economy benefits from a market that prices infrastructure quality, the same way the cloud economy is priced by reliability.

Standby capacity and graceful failover

Each cluster has a standby of three operators in addition to its seven active members. A standby operator participates in the cluster's communications, is listed on the cluster roster with its own independent ML-DSA-65 key, and is ready to step in when an active operator goes offline or is rotated out.

The standby surface is part of what makes the cluster model **operationally resilient** rather than just structurally redundant. An active operator can be removed (by the cluster, by network policy, or by their own departure) without any ceremony or key handover, because every operator already signs with the independent key it alone holds and a valid certificate needs only seven of the cluster's ten members' signatures. When an active operator drops, an online standby's signature fills the 7-of-10 threshold automatically — failover is free, the threshold continues to hold, and the cluster keeps producing.

Multi-cluster operator caps

A single operator entity cannot dominate the cluster set. Operator identity is verified across all cluster memberships, and a hard cap limits how many active cluster positions any single operator can hold. This prevents a successful operator from accumulating positions across many clusters and reproducing single-operator centralisation on top of the cluster substrate. Service-tier specialization (GPU, RPC, archive, oracle) is encouraged across clusters; consensus-position multiplicity is bounded.

The combination — public cluster composition + standby capacity + per-operator multi-cluster caps + per-wallet delegation caps (§16) — produces a network that does not naturally concentrate, even under sustained success of any single operator or cluster.

7. Post-Quantum from Genesis

Monolythium uses post-quantum cryptography as its **default** at the user-facing signature layer. ML-DSA-65 (NIST FIPS 204, also known as Dilithium Level 3) is the only signature primitive accepted at transaction admission. There is no classical fallback, no hybrid mode, no ECDSA acceptance path “for compatibility.” Every account signs every transaction with a post-quantum primitive from the chain’s first block.

This is unusual. Most production EVM-compatible chains use classical secp256k1 or Ed25519. Monolythium requires mandatory post-quantum user signatures while still supporting the cross-chain interop edge that EVM-resident value lives behind.

It is also a deliberately understated headline. **Post-quantum is a property of the chain, not its purpose.** The chain exists to settle the autonomous economy; post-quantum cryptography is one of several properties that make that settlement layer trustworthy on a multi-decade horizon. The audience the chain serves needs identity that remains unforgeable for the lifetime of an agent’s reputation — not because “quantum-proof” sells, but because that is the audience’s real requirement.

The whole stack, not just signatures

Cryptographic posture is end-to-end:

LAYER	PRIMITIVE	USE
User signatures	ML-DSA-65 (FIPS 204)	Every user-signed transaction
Consensus signatures	ML-DSA-65 (FIPS 204)	Per-operator vertex signing; cluster quorum = 7-of-10 bitmap multisig of independent operator signatures
Emergency recovery (removed)	None at launch — ML-DSA-65-only	The SLH-DSA hash-based backup key, the emergency-key registry, and in-protocol algorithm rotation are removed ; a cross-family backup + rotation is a future capability for a subsequent genesis (reconciliation note, item 10)
Key encapsulation	ML-KEM (FIPS 203)	Peer-to-peer handshakes (§12.3), RPC TLS, stealth-address derivation
Zero-knowledge verification (gated off)	Post-quantum FRI/STARK — intended direction	Application-layer proof verification is disabled at genesis . The earlier SP1 zkVM + Groth16-BN254 verifier is not enabled; the direction is a post-quantum FRI/STARK verifier that ships gated off until ready. Consensus finality is pure ML-DSA-65 and never depended on it
Hash	BLAKE3	State-tree leaves, Merkle commitments, content-addressed proofs, address derivation

Three properties of this stack are worth highlighting because they reflect design choices rather than ingredient-list ticks:

- **No hybrid signature mode.** The chain does not accept “signed with ECDSA AND ML-DSA-65”; it accepts only ML-DSA-65. The reasoning is below.
- **Post-quantum at consensus, not just at the user layer.** Every anchor is finalized by post-quantum signatures. Each operator in a cluster signs the cluster’s vertex with its own ML-DSA-65 key, and an anchor is final once a 7-of-10 quorum of those per-operator signatures is present, gathered across a $2f+1$ count-based quorum of clusters — clusters carry equal weight, and stake sets only top-100 admission rank, not vote weight. There is no shared key, no aggregate, and no classical signature anywhere in the consensus path. The consensus layer is single-tier and fully post-quantum.
- **The honest cost of post-quantum consensus.** ML-DSA-65 signatures are large and do not aggregate. A cluster’s 7-of-10 quorum is a bundle of seven raw 3,309-byte signatures (about 23 KB per cluster), and an anchor certificate concatenates one such bundle per voting cluster, so the certificate grows linearly with the number of voting clusters. This is a deliberate trade: the chain pays real per-anchor bandwidth

in exchange for post-quantum, stateless finality with no classical signature anywhere in the consensus path. Section 12.4 states the exact byte cost.

Why pure post-quantum, not hybrid

The temptation in this transition era is to ship “hybrid” mode — a transaction carrying both an ECDSA and a post-quantum signature. Hybrid comes in two shapes that behave very differently: consensus can accept **either** signature, or it can require **both**. Hybrid is appealing because it preserves user-experience continuity and offers theoretical defense in depth.

Monolithium rejects hybrid for three reasons.

First, the defense-in-depth argument does not survive either shape.

If consensus accepts **either** signature, the scheme is only as strong as its **weakest** component. An adversary able to forge ECDSA on a sufficiently large quantum computer simply presents an ECDSA-signed transaction; the post-quantum half is never examined. That is not defense in depth — it is a standing downgrade path, and it defeats precisely the threat the post-quantum primitive was added to address.

If consensus requires **both**, that attack fails: a quantum adversary who forges the ECDSA half still cannot produce an ML-DSA-65 signature over the new message. But then the post-quantum primitive is already carrying the entire quantum defense on its own, and the ECDSA half is not hedging the quantum threat at all. What it hedges is the opposite risk — a classical break of the lattice assumption. That risk is real and worth naming plainly: ML-DSA-65 is younger than ECDSA and has absorbed less cryptanalysis. Monolithium’s answer to it is process rather than a permanent second signature on every transaction — the emergency freeze (§23.2) pauses admission while a primitive migration is coordinated, as described under “The quantum hedge” below. A standing classical path would buy that hedge at the price of two code paths, two keys per account, and a second verifier maintained forever, against an event that would be visible in the literature well before it was exploitable.

Second, hybrid creates audit ambiguity. Two acceptance paths means two code paths. Two code paths means twice the surface for implementation bugs, twice the surface for downgrade attacks (an adversary suppresses the post-quantum half and forces verification through the classical path), and twice the verifier complexity. With one signature primitive, the verifier does one thing; the audit story is “ML-DSA-65 verification is correct” rather than “the verifier correctly handles three combinations of two signature types.” This is a meaningful simplification.

Third, hybrid postpones the migration. A chain that ships hybrid commits to the migration twice — once now (to add the post-quantum primitive) and once later (to remove the classical primitive). The migration later is harder, because by then ECDSA has been deprecated for years and tooling has accumulated around the assumption that classical signatures work. Monolithium does the migration once, at genesis, when there is no installed base of classical-only wallets to support.

The cost of pure post-quantum is signature size — ML-DSA-65 signatures are roughly 3,309 bytes versus ECDSA's 64 bytes. The cost is real and is paid in storage and bandwidth. The alternative is paying the migration cost twice.

The quantum hedge

Post-quantum cryptography is not unconditionally future-safe. Cryptographic primitives have historically had finite useful lifetimes, and lattice-based cryptography is no different. The honest position is that ML-DSA-65 is the right choice for the foreseeable horizon, that it will likely remain a good choice for a decade or more, and that a future cryptographic break should be planned for rather than denied.

Reconciliation (v6, item 10) — crypto agility is a future capability, not a live feature. An earlier design planned for a break through an in-protocol mechanism: an **emergency-key registry** in which a user pre-registered a hash-based **SLH-DSA (FIPS 205)** backup key alongside their ML-DSA-65 key, plus an **algorithm-rotation** event that would refuse the broken primitive at a set anchor height and move registered users to their backup. That mechanism was found inoperative in audit and has been **removed**: the chain is **ML-DSA-65-only**, with no second-family backup key and no in-protocol algorithm swap at launch.

The honest present-tense posture is that a future ML-DSA-65 break is planned for at the level of **process**, not an armed on-chain switch: the emergency **freeze** circuit breaker (§23.2) can pause transaction admission while a signature-primitive migration is coordinated as a maintainer-and-operator hard fork. Post-quantum **crypto agility** — reintroducing a cross-family backup primitive and an in-protocol rotation path so that a break-day migration is automatic rather than coordinated — is a capability the project intends to bring back **in a subsequent genesis**, and it is described here as future, not live. Hash-based signatures rest on different mathematical assumptions than lattice-based ones, so a cross-family backup remains the right shape for that future mechanism; what changed is that it is no longer present on the chain today.

8. Forking as the Exit Path

The chain refuses on-chain governance (§4.1). The replacement model needs an answer to the question: what happens if a significant fraction of holders, operators, or users disagree with the chain's direction?

The answer is **forking is legitimate**. Not as a threat, not as a last resort, but as the explicitly documented path of dissent.

The chain's text and source code are released under licenses that permit forking. Operators are free to run alternative software. The cluster substrate, the cryptographic primitives, the SDK, the explorer, and the wallet are all open enough for an alternative project to take Monolithium's design and run with it under different leadership, different parameter choices, or a different philosophy.

This is the inverse of the "governance capture" failure mode. A protocol that uses on-chain governance to absorb dissent must keep losing capture battles to remain credibly neutral; a protocol that says "if you disagree, fork" makes its neutrality the absence of a capture surface. There is nothing to capture, because direction is set by maintainers and operators, and exit is set by anyone with a compiler.

The point is not that the network expects forks; the point is that the right to fork is the structural guarantee against capture, and it is the only one the chain offers — but it is real.

9. Why Rust and RISC-V

The execution layer is Rust-first smart contracts compiled to a deterministic RISC-V execution target. There is no Solidity, no EVM bytecode, no Ethereum precompile model in the mainnet execution path.

EVM compatibility is valuable because it answers many practical market questions at once. When a developer asks whether a new chain is EVM-compatible, they usually mean:

- Can I bring assets over?
- Can I bridge major stablecoins, ETH, BTC, and other large assets?
- Can my wallet display the assets and route risk visibly?
- Can I create and trade tokens?
- Can my application reach users without rebuilding every surrounding integration?

EVM-compatibility answers those by inheriting the Ethereum tooling surface: token standards, contracts, wallets, explorers, auditors, libraries, and liquidity routers.

Monolythium answers them differently. The **execution** layer stays Rust/RISC-V-native; the **liquidity** layer connects outward through an external interop provider and issuer-supported integrations, rather than an in-tree cross-chain bridge. The chain avoids isolation while keeping the base layer smaller and cleaner.

What Monolythium gives up by not being EVM-compatible

Real costs:

- existing EVM bytecode does not run;
- Solidity is not the default developer model;
- Hardhat, Foundry, OpenZeppelin, the Uniswap fork lineage, and proxy-pattern ecosystems do not carry over directly;
- some Solidity-focused developers and market makers will ignore the chain;
- Mono must build its own SDK, ABI, wallet UX, explorer support, token standards, and testing tooling.

This is paid intentionally.

What Monolythium gets

The upside is a stronger identity and structural advantages.

- **Rust** provides strong compiler feedback, clear state-machine design, ergonomic testing and fuzzing, and better alignment with AI-assisted development than Solidity. Rust does not make contracts magically safe — poor economics, bad access

control, oracle mistakes, bridge bugs, and unsafe host interfaces can still cause serious failures — but it removes entire classes of avoidable bugs, makes contracts easier to reason about, and reduces the amount of custom code needed for common financial primitives.

- **RISC-V** is open, simple, portable, deterministic, and increasingly aligned with the zero-knowledge proving ecosystem. It gives Monolythium a clean execution target without inventing a custom bytecode language. Tooling, debuggers, formal-verification work, and zero-knowledge provers all converge on RISC-V; the chain rides that convergence rather than fighting it.
- **Native modules.** Common financial primitives can be implemented and audited once at the protocol level. Agent spending-policy checks remain a target §18 module and may be described as native only after their separately versioned activation.
- **AI-assisted development.** AI coding assistants are dramatically more useful in Rust than in Solidity. The Rust ecosystem has fifteen years of training data; Solidity does not. The next decade of contract development will lean heavily on AI assistance, and Rust is structurally a better target language for it.
- **Cleaner audit surface.** Each native module is audited at the protocol level. Applications that compose on top inherit that audit work. The chain pushes the heaviest cryptographic and financial logic into the well-audited core and leaves applications a smaller, simpler API to compose against.

The chain has chosen a harder adoption curve in exchange for a cleaner identity, a stronger long-term foundation, and a better fit for the workloads it expects to serve. EVM chains compete on liquidity gravity. Monolythium competes on a different axis: settlement for the next decade's economic actors, on rails that are not borrowed from the previous decade.

10. Market Positioning

Monolythium does not have a perfect mirror competitor.

- Ethereum and EVM-compatible chains compete on liquidity and developer familiarity.
- High-performance non-EVM chains compete on throughput and consumer applications.
- High-performance trading chains compete for low-latency derivatives flow.
- Open agent-payment standards (x402, AP2, ACP, MCP) compete for the payment handshake; Monolythium's unactivated target architecture aims to compose underneath compatible released integrations rather than compete at that handshake (§3).

Monolythium's differentiated combination is:

- AI-agent settlement as a primary category;
- Rust/RISC-V-native execution from the base layer;
- post-quantum accounts as default, not optional;
- native MRC token, NFT, and market modules, plus a target agent-commerce architecture;
- external interop-provider liquidity rather than EVM execution compatibility;
- no on-chain governance;
- no mainnet perpetuals;
- structurally non-fungible public/private denomination;
- a public cluster marketplace with distributed validator technology;
- a lean, post-quantum core that keeps the highest-risk surfaces out of consensus — cross-chain interop runs through an external provider, and application-layer zero-knowledge verification ships gated off pending a post-quantum FRI/STARK verifier rather than a live Groth16 verifier;
- a target path to composition with compatible agent-payment standards after the required protocol, SDK, wallet, asset, and connector releases exist.

If the market only wants another EVM chain, Monolythium has chosen the harder path. If the market wants safer open settlement for agents, payments, spot markets, and long-lived digital identities, the chain has a distinct position. The bet is not that EVM disappears — it will not, for at least a decade. The bet is that the settlement layer for the next decade's most interesting workloads will not be a copy of the chain that hosts the last decade's.

Why large participants would use a shared network

A large company can build infrastructure. That does not automatically make its infrastructure a neutral settlement layer.

For agent commerce, neutrality matters because agents transact across companies, model providers, wallets, jurisdictions, and service providers. A payment rail owned by one AI lab or one platform may work inside that platform, but it is less credible as a universal settlement substrate. A bank or a fintech that wants its agent flow to interoperate with the rest of the agent economy needs a substrate that is not owned by a competitor.

Monolythium's current value proposition to large participants is shared settlement rules, verifiable state, common market standards, and open infrastructure participation. Portable agent identity, portable reputation, spending policy, and agent settlement are unactivated vNext objectives that require the §18 release stack. Large participants may still run clusters, build applications, issue assets, operate provers, or provide liquidity. The intended benefit is a shared network rather than fragmented private ledgers, without presenting the gated agent layer as delivered.

PART 2 — HOW MONOLITHIUM WORKS

11. Consensus: Starfish-C

Monolithium's consensus engine is **Starfish-C**, a DAG-BFT protocol derived from the Starfish family of distributed-acyclic-graph consensus designs. Starfish-C provides deterministic linearization of a directed-acyclic graph of cluster-signed vertices, **four-second deterministic finality** under the partial-synchrony assumption (four to eight seconds typical, with an outer bound of about twelve seconds under degraded networking before a view-change is triggered), and equivocation handling that produces succinct, on-chain proofs and a one-hundred-percent slash plus permanent operator exile on detection.

The four-second cadence is chosen to leave headroom for cluster-marketplace geographic diversity (operators on any continent participate cleanly) and to allow consensus signature collection under load. The chain optimizes for steady performance under adversarial conditions, not racetrack speed under ideal conditions.

Safety

Safety is guaranteed through **deterministic linearization** of the DAG. Three properties combine.

No forks. Clusters sign exactly one vertex per round. A vertex is valid only if it references a quorum ($2f+1$, where f is the Byzantine bound) of vertices from the prior round. Two vertices signed by the same operator at the same round constitute a slashable equivocation.

The causal cone. When a wave leader is committed, the leader's entire causal history — the recursive set of vertices the leader's vertex transitively references — is committed in a deterministic linear sequence. The linearizer is a pure function of the DAG state; two honest clusters starting from the same DAG state derive byte-identical block sequences.

Bounded reorg. The maximum reorg depth is the size of the causal cone for an unfinalized leader, bounded by protocol parameter. An adversary cannot cause a reorg deeper than this bound without controlling more than f Byzantine clusters, which is excluded by the Byzantine threshold assumption.

Liveness

Liveness holds under partial synchrony — the standard BFT assumption that messages are eventually delivered within an unknown but finite bound. Starfish-C achieves liveness through wave-based progression: time is divided into waves, each wave selects a leader whose committed vertex anchors the linearization of its causal cone, and if a

wave leader fails to be committed within its timeout, the linearizer executes a skip round and the next successful leader's commit covers the skipped wave. Malicious subsets cannot stall the network as long as a quorum ($2f+1$) of honest clusters is active.

Leader selection

Wave leadership is assigned by a **deterministic round-robin schedule** over the active cluster set: the leader of a wave is the $(\text{wave number} \bmod \text{active-cluster-count})$ -th cluster in the canonical active ordering. Every honest cluster computes the same leader from the same public DAG state — the wave number and the active set — with no seed, no beacon, and no separate leader-selection signature. This is the pacemaker schedule the Starfish family specifies: leadership rotates predictably, so liveness depends only on eventually reaching an honest leader rather than on agreeing a shared random beacon.

Round-robin removes a class of grinding and seed-manipulation risk by construction: there is no seed for an adversary to bias by withholding or selectively releasing a partial signature, because there is no aggregate seed at all. Leadership is a schedule, not a lottery. If a scheduled leader fails to be committed within its timeout, the linearizer executes a skip round and the next scheduled leader's commit covers the skipped wave (see Liveness, above).

Anti-equivocation

Double-signing — an operator signing two distinct vertices at the same round — triggers a **succinct equivocation proof** (the two conflicting signed vertices, both verified against the same operator's signature key) and a **100% slash of the operator's self-bond**. The operator is permanently exiled from the cluster set. Recovery by re-bonding is not permitted; the operator identity is burned.

The slash is exemplary, not just punitive. A protocol that tolerates equivocation invites it. A protocol that destroys the equivocating operator's stake and bars them forever does not need to tolerate it.

Anchors, blocks, and waves

The user-facing unit on Monolythium is the **anchor** — the canonical, finalized point in the chain history at which state transitions become irreversible. Anchors are produced from waves through deterministic linearization of the DAG's causal cone. Multiple internal layers exist:

- **Vertex.** A single cluster's signed round payload.
- **Wave.** A round of vertex production across the cluster set.
- **Anchor.** A deterministic linearization point in the wave structure; the user-facing unit of finality.

- **Block.** The traditional name preserved for compatibility with EVM-style RPC fields (such as `eth_blockNumber`) where an external tool expects a height-like integer. The block sequence is an aliased view of the anchor sequence.

Wallets, explorers, and the chain's SDK use **anchor** as the canonical term. A skipped wave produces zero anchors; an anchor that is committed advances the anchor height. The aliased block view exists for cross-chain tooling that needs a familiar integer label.

Performance against attack

Two properties matter most under adversarial load.

Chain-bomb resilience. A chain-bomb attack — an adversary attempting to flood the DAG with malformed or out-of-order vertices to inflate end-to-end latency — is bounded by a forward-clock cap parameter. Vertices more than N rounds in the future relative to the receiver's local wave clock are dropped at the gossip layer.

Payload efficiency. The amortized per-round payload cost scales linearly in the cluster count via Reed-Solomon shard dissemination, rather than quadratically as in earlier DAG-BFT designs. The result is a protocol that scales to substantially larger cluster sets than the current target without an architectural rewrite.

What this gives users

For a user or a builder, the important outcomes are:

- transactions finalize under a defined, mathematically grounded consensus process;
 - anchor checkpoints can be verified;
 - external interop relayers and indexers can rely on canonical state commitments;
 - operators have clear responsibilities and incentives;
 - adversarial conditions degrade the chain's latency, not its safety.
-

12. Cryptography

This section specifies the cryptographic primitive set in implementation detail. Section 7 covers the design philosophy.

12.1 Primitive set

LAYER	PRIMITIVE	USE
User signatures	ML-DSA-65 (FIPS 204, Dilithium Level 3)	Every user-signed transaction
Consensus signatures	ML-DSA-65 (FIPS 204, Dilithium Level 3)	Per-operator vertex signing; cluster quorum = 7-of-10 bitmap multisig of independent operator signatures
Emergency backup signatures (removed)	None at launch — ML-DSA-65-only	The SLH-DSA hash-based backup and the emergency-key registry are removed ; a cross-family backup + in-protocol rotation is a future-genesis capability (§23.1; reconciliation note item 10)
Key encapsulation	ML-KEM-768 (FIPS 203, Module-Lattice KEM)	Peer-to-peer handshakes (a KEM-Noise IK-style construction of our own, pending cryptographer review); RPC TLS; stealth-address derivation
Zero-knowledge verification (gated off)	Post-quantum FRI/STARK — intended direction	Application-layer proof verification is disabled at genesis ; the earlier SP1 zkVM + Groth16-BN254 verifier is not enabled (§12.6). Consensus finality is pure ML-DSA-65 and never depended on it
Hash	BLAKE3	State-tree leaves, Merkle commitments, content-addressed proofs, address derivation

There is no Ed25519 acceptance path. There is no hybrid signature mode. The protocol validates exactly one signature primitive at transaction admission: ML-DSA-65. The earlier SLH-DSA emergency-rotation alternative has been **removed** (§23.1); there is no second signature primitive on the chain today.

12.2 ML-DSA-65 (Dilithium Level 3)

Standard. NIST FIPS 204, parameter set 3 — security comparable to AES-192.

Security basis. Module-Learning-With-Errors (M-LWE) hardness. M-LWE is the lattice problem on which Dilithium rests. No known quantum or classical algorithm breaks M-LWE in feasible time at the parameters used.

Performance. Verification on commodity hardware completes in ~0.12 ms per signature; signing in ~0.18 ms. Signature size is 3,309 bytes. Public-key size is 1,952 bytes. Private-key size is 4,032 bytes.

Determinism. Signing uses the deterministic variant of Dilithium with no random nonce. This eliminates the class of vulnerabilities arising from weak-random-source signing.

12.3 ML-KEM (Module-Lattice KEM)

ML-KEM-768 is used wherever a key encapsulation is needed:

- **Peer-to-peer handshakes.** Operator-to-operator connections use ML-KEM-768 for a KEM-Noise **IK-style** mutual handshake: two ML-KEM-768 encapsulations — one under each side’s long-lived static key — feeding an HKDF-SHA-256 key schedule bound to a transcript hash, with an HMAC-SHA-256 transcript MAC and a ChaCha20-Poly1305 record layer. It replaces the classical X25519 ECDH of libp2p’s stock Noise upgrade, which is bypassed entirely; there is no hybrid stage and no classical fallback.

This is our own construction, and it has not yet had an independent cryptographic review. It is not a stock Noise pattern: libp2p’s Noise implementation hardcodes X25519 with no plug-point for a different key exchange, so rather than fork it we ship a self-contained handshake satisfying the same libp2p upgrade contract. The KEM-Noise key schedule, the transcript-MAC construction, and the nonce discipline are composed by us. A senior-cryptographer review of the wire protocol and its test vectors is owed before mainnet and is tracked as a release gate; until it closes, the running testnet’s transport should be read as unreviewed.

Two classical artefacts remain at this layer, and neither authenticates consensus or state. `x25519-dalek` is still linked through libp2p’s Noise crate but is never invoked — the handshake slot is occupied by the post-quantum upgrade — and an `ed25519` keypair is retained solely to derive the transport-layer `PeerId`, whose format libp2p hardcodes. - **RPC TLS.** Public RPC endpoints serve TLS certificates with ML-KEM-encapsulated session keys. Wallet, indexer, and explorer connections to nodes use post-quantum TLS. - **Stealth addresses.** The privacy denomination’s stealth-address scheme uses ML-KEM end-to-end for one-time-address derivation.

12.4 Single-tier post-quantum finality

Consensus uses a **single** finality tier, and it is fully post-quantum. There is no fast classical tier and no separate quantum-attested checkpoint: every anchor is finalized the same way, by post-quantum signatures, the moment it commits.

How an anchor is finalized. Each operator in a cluster signs the cluster’s vertex with its own ML-DSA-65 key. A cluster’s quorum certificate is a **7-of-10 bitmap multisig**: a one-byte-resolution bitmap of which operators signed, followed by the raw per-operator ML-DSA-65 signatures themselves. An anchor is final once a $2f+1$ count-based quorum of clusters — each carrying equal weight, with stake setting only top-100 admission rank and never vote weight — has contributed a valid 7-of-10 cluster vote. Anchor-level finality settles in roughly four to eight seconds (one wave at the four-second round duration). The same certificate is the everyday-transfer signal and the deep-settlement signal that exchange deposits, external interop attestations, and other high-value cross-chain settlement bind to; there is one finality, post-quantum at every depth.

The honest bandwidth cost. ML-DSA-65 signatures are 3,309 bytes each and do not aggregate. Concretely, on the current testnet, per cluster (ten operators, threshold seven; the live fleet runs two such clusters — see the top-of-file reconciliation note):

- A single 7-of-10 cluster vote is **23,229 bytes**: seven raw 3,309-byte signatures plus a few bytes of length framing and the operator bitmap.
- A full anchor quorum certificate over that one cluster is **23,301 bytes**.

This is the deliberate trade. An anchor certificate is a bundle of raw per-operator ML-DSA-65 signatures that grows linearly with the number of voting clusters. In exchange it gets post-quantum, stateless finality with no signature aggregation, no shared key, and no interactive key-setup ceremony of any kind — each operator simply signs with its own key — and no classical signature anywhere in the consensus path for a future quantum adversary to forge. The cost is real per-anchor bytes; the benefit is that there is nothing classical left to break at consensus.

At scale. The cost is perfectly linear: each added voting cluster contributes another ~23.2 KB to the certificate. At the whitepaper’s target of roughly 100 clusters with a $2f+1$ (67-of-100) quorum, one quorum certificate is about 1.5 MiB; if all 100 clusters vote, about 2.2 MiB. Each anchor carries a small fixed number of such certificates (round, leader, and data-availability quorum certificates). This bandwidth is the price of the post-quantum posture, sized honestly here so operators can provision for it; it is not hidden behind a “tiny aggregate” claim.

12.5 Mempool

The mempool is plaintext: a transaction body is visible from the moment it enters the mempool until anchor inclusion, and the protocol does not seal or encrypt mempool contents. Transaction-ordering fairness is addressed at the DAG-consensus layer — ordering follows the deterministic DAG linearization rather than a single sequencer’s discretion, so a plaintext order is sequenced by consensus rules rather than by privileged pre-inclusion reads.

The machine-checked `no_classical_in_protocol` lint is enforced as a hard build fail: the consensus and signature/KEM protocol path carries no classical asymmetric primitive. **No signature or KEM carve-out is allow-listed.** The classical primitives — secp256k1, Ed25519, WebAuthn P-256 — the crate that hosted them, and the classical/

post-quantum hybrid construction were removed from the tree outright, and their wire tags are permanently retired: a transaction presenting one resolves to no algorithm and is rejected before verification. The lint's only remaining exemptions are the zero-knowledge proof-verifier boundary (§12.6), which is neither a consensus signature nor a KEM and is disabled at genesis, and the peer-to-peer transport's node-identity keys (§12.3), which authenticate network connections rather than consensus or state. The precise claim is therefore **no classical primitive in the protocol path**, not “no classical code in the dependency graph.” The earlier confidential-transfer exemption (the privacy precompile's Ristretto/Pedersen group arithmetic) is gone because that construction itself was removed (§19).

12.6 Zero-knowledge proof systems

Application-layer zero-knowledge verification is **disabled at genesis and is not reachable in the public v0.4.0 protocol**. It is not consumed by the consensus path and never was: consensus finality is pure ML-DSA-65 (§12.4), and no proof system settles state. The earlier design carried an SP1 zkVM with a Groth16-BN254 verifier, but that verifier is not enabled. The intended direction is a hash-only, fully post-quantum **FRI/STARK verifier**, gated until the proving ecosystem delivers on-chain-verifiable STARK receipts.

When an application-layer verifier is eventually armed, the roadmap targets zero knowledge where it reduces risk the most. Each of the following is a **gated, future capability, not a live one**:

- **Swap proofs.** A swap could verify that a batch of intents or orders was matched according to a declared policy — supporting fair ordering, batch auctions, and verified-matching markets without asking users to trust an opaque sequencer.
- **zkML attestations.** A model could produce a verifiable attestation that an output was generated by a specific signed model on specific inputs.
- **High-value off-chain computation.** Computations that would be prohibitive to run on-chain could be proven off-chain and verified on-chain in bounded gas.

Cross-chain interop no longer uses an in-tree bridge verifier: the in-tree bridge proof stack was removed (§20), and the chain does not verify bridge proofs on-chain. The GPU prover service tier — described in §17 — is likewise **gated off at genesis**; when armed it will provide paid proof-generation capacity while on-chain verification runs on commodity-CPU clusters, so that generation stays concentrated on operators with the right hardware while verification stays cheap and distributed. None of it is currently reachable.

12.7 Hash functions

BLAKE3 is the chain's primary hash for state-tree commitments, content-addressed proofs, and public address derivation. Grover's quadratic speedup against 256-bit output yields 128-bit quantum security, well above any quantum threat horizon. Where legacy compatibility codecs require Keccak (such as event-log selectors), that use is explicitly scoped to compatibility, not to address derivation or state commitments.

13. Identity: Addresses, Mnemonics, and Names

13.1 Account addresses

Monolythium addresses are 20-byte payloads derived from a domain-separated BLAKE3 hash of the user's public key:

```
address[20] = BLAKE3(  
    "MONO_ADDRESS_BLAKE3_20_V1"  
    || algo_id_be_u16  
    || canonical_pubkey_bytes  
)[0..20]
```

The wallet publishes the user's public key on chain at account creation, providing post-quantum forgery resistance from the moment the account exists, before any funds arrive. Until publication, the account is identified only by its address; after publication, every transaction is verified against the stored public key.

13.2 Bech32m display

User-facing addresses are displayed in **bech32m** format only. Hex `0x...` display is not the canonical surface; bech32m is the only address format wallets, the chain's block explorer, partner integrations, and SDKs render by default.

The bech32m checksum and the `mono` human-readable prefix make Monolythium addresses visually unmistakable from any other chain's `0x...` format, providing format-as-safety against cross-chain address confusion at exchange listings, bridge UIs, and partner integrations.

Addresses use a per-type human-readable prefix discriminator so that an address's role is visible at a glance:

PREFIX	ADDRESS CLASS
<code>mono1...</code>	Standard user account
<code>monos1...</code>	Stake / delegation account
<code>monoc1...</code>	Contract account
<code>monok1...</code>	Cluster account
<code>monom1...</code>	Native module account
<code>monox1...</code>	External-interop (bridge) account

Additional prefixes are reserved for future account classes. A user reading the discriminator can tell whether they are about to send to a normal account, a contract, a cluster, or an external-interop endpoint — a property that hex addresses cannot offer.

13.3 Wallet recovery — PQM-1 removed

PQM-1 is removed and its old algorithm-tagged seed layout must not be implemented. The canonical mnemonic input is a standard 24-word BIP-39 phrase, re-derived for ML-DSA-65 as:

```
mldsa_seed = SHAKE256(
    "monolythium.mldsa65.v1" || bip39_pbkdf2_seed(mnemonic, ""),
    32
)
```

The browser wallet encrypts its local vault with Argon2id and XChaCha20-Poly1305 and stores the encrypted record in extension-local storage. That is an application vault, not a generic operating-system keychain or biometric guarantee. Selecting a network does not change key derivation. Recovery compatibility with another client—especially the desktop wallet—must be demonstrated by an explicit cross-client recovery test before users are told the clients are interchangeable.

Users must never enter a mnemonic into Steele, hosted MCP, documentation search, support chat, or any web service. Steele authenticates through the browser wallet and does not receive wallet secrets.

13.4 The LYTH name registry

Bech32m addresses are unmistakable but not memorable. Monolythium ships a hierarchical, on-chain **name registry** that maps human-readable names to addresses.

Naming structure:

- **Charset.** Lowercase ASCII letters, digits, and the hyphen. No mixed case, no Unicode confusables, no scripted characters at the protocol level.
- **Hierarchy.** Names live under the `.mono` namespace and resolve to structural categories. A bare `<name>.mono` is a **human** (personal) account. The unactivated vNext agent design reserves `<name>.agent.<human>.mono` for an agent beneath its principal; this is a target namespace, not evidence that agent sub-accounts or their registration ABI are live. Clusters, deployed contracts, and protocol/system entities use their released categories; any closed `system` namespace remains subject to the activated registry contract.
- **Pricing.** A U-curve pricing schedule discourages both squatting (short, premium names cost meaningfully more) and clutter (very long names cost a small

administrative fee). The pricing curve is constitutional; it is not adjustable through any signaling mechanism.

- **Transfer.** Names transfer through a propose-accept flow with a 24-hour acceptance window. The recipient must affirmatively accept the transfer; the sender cannot push a name to an unwilling recipient.
- **Reserved prefixes.** The registry refuses to register names that begin with the chain's bech32m human-readable prefixes (`mono`, `monos`, `monoc`, `monok`, `monom`, `monox`, and additional reserved prefixes), or with `0x`. This prevents the registry from minting names that could be confused with raw addresses.
- **Cascade target.** A future agent-identity transfer must define deterministic handling for subordinate names so they cannot become orphaned silently. No current agent-transfer behavior is asserted.

The Monolythium Foundation seeds the registry at genesis with a reserve of category-defining and protocol-relevant names. The reserve is published, not sold; reserved names are released into the open market on a schedule that is also published.

Illustratively, a person could be addressed as `alex-rivera.mono`, while the vNext target would represent an agent as `support-bot.agent.alex-rivera.mono`. Cluster, contract, and system examples depend on the activated registry categories. The bech32m address remains the canonical underlying identifier; a released name is only a human-readable alias that resolves to it.

14. Execution: Rust on RISC-V

The execution environment is a **Rust/RISC-V-native smart contract runtime**. The chain does not target Solidity or EVM bytecode for mainnet. The execution layer has three tiers:

1. **Native protocol modules** for activated hot and security-critical primitives. Agent-commerce registries and spending-policy enforcement are unactivated vNext targets and are not members of the current public capability set.
2. **Rust/RISC-V contracts** for application-specific programmable logic.
3. **zkVM-proven computation** (gated off at genesis — see §12.6) — a future application-layer verifier for high-cost off-chain verification and, when armed, verified-matching swaps and zkML attestations. It is not reachable today, and cross-chain interop is handled by an external provider (§20) rather than by in-tree bridge proofs.

The goal is not “EVM, but faster.” The goal is a smaller, safer programmable settlement layer with first-class support for post-quantum accounts, AI-agent workflows, and clean external-provider cross-chain settlement.

14.1 Contract artifact

The contract artifact is a deterministic Monolythium RISC-V package containing:

- code section compiled to the approved RISC-V profile;
- contract ABI manifest;
- import table declaring required host syscalls;
- declared memory limits;
- declared storage namespace;
- code hash and build metadata;
- optional source map and debug metadata, excluded from consensus hashing.

The preferred source language is Rust. The contract SDK provides macros, types, storage collections, ABI generation, event helpers, test harnesses, and deployment packaging.

Contract execution is fully deterministic. There is no wall clock, no filesystem, no sockets, no host randomness, no floating-point. Every observable input arrives through transaction calldata, contract storage, anchor context exposed by syscall, or explicit native-module query.

14.2 Host syscall ABI

The RISC-V VM is intentionally small. Chain access happens through a narrow host syscall ABI:

```
storage_read(key) -> bytes?
storage_write(key, value)
storage_delete(key)
caller() -> Address
contract_address() -> Address
anchor_height() -> u64
anchor_hash(height) -> Hash
call_contract(address, input, value) -> bytes
emit_event(topic, data)
transfer_native(to, amount)
verify_signature(algo, pubkey, msg, sig) -> bool
hash(domain, bytes) -> Hash
revert(code, data)
```

Syscalls are metered separately from RISC-V instruction cycles. Storage reads and writes are priced against state I/O, not CPU. Cryptographic syscalls are priced against measured native implementation cost. Cross-contract calls create a bounded call frame with explicit gas/cycle budget, memory limit, value transfer rules, and revert propagation.

The syscall ABI is a **consensus contract**. Once a syscall reaches stable status, changing its semantics requires a coordinated upgrade.

14.3 Gas, cycles, and state

The cost model separates three resources:

RESOURCE	WHAT IT MEASURES
RISC-V cycles	Deterministic compute performed by a contract
Host syscall cost	Native functions such as hashing, signature checks, transfers, and module calls
State I/O	Reads, writes, deletes, storage growth, events, and receipts

This matters because **state is the real bottleneck** in a blockchain. Arithmetic is cheap. Long-term storage, database reads, and permanent state growth are expensive. Monolithium prices state intentionally so normal usage stays affordable while state bloat remains controlled.

The practical result:

- simple transfers are cheap;

- token and NFT operations are predictable;
 - compute-heavy contracts benefit from efficient RISC-V execution;
 - storage-heavy applications pay for the burden they place on the network;
 - wallets can estimate fees more accurately because execution is more predictable than under EVM.
-

15. Native Modules and MRC Standards

Monolythium does not remove tokens, NFTs, or markets. It makes them **native**.

Instead of asking every application to rebuild core financial primitives as arbitrary smart contracts, the chain provides audited native modules and Mono-native standards.

ETHEREUM CONVENTION	MONOLYTHIUM TARGET
ERC-20	MRC-20 fungible tokens
ERC-721	MRC-721 non-fungible tokens
ERC-1155	MRC-1155 multi-assets
ERC-4626	MRC-4626 vaults
ERC-1271 / ERC-4337 patterns	Native smart-account and policy-account standards
Solidity marketplace contracts	Native marketplace modules and Rust/RISC-V contracts
Contract-based AMMs and order books	Native spot-market modules with contract hooks

The high-volume paths are native:

- token transfers;
- NFT ownership and transfer;
- multi-asset and game-item batch transfers;
- spot-market order placement and settlement;
- agent spending-policy checks (target; not activated);
- account and permission management.

Rust/RISC-V contracts remain available for application-specific logic: custom marketplaces, vault strategies, agent workflows, escrow templates, games, domain-specific settlement flows, and advanced business logic.

The design reduces duplication and audit burden. A common asset standard should not be reimplemented thousands of times across unrelated applications. Wallets and explorers display MRC assets as **first-class assets**, not as arbitrary contract calls — when a user receives an MRC-721, the wallet knows it is an NFT with a typed schema, not an opaque series of bytes that might or might not implement an interface correctly.

Markets and NFTs

The native order book is intended as general market infrastructure for released assets and markets. Agent-service settlement is an unactivated vNext use case, not a current order-book or Steele capability. Perpetuals and margin remain explicitly removed (§4.2). Automated market makers can be implemented either as native modules or Rust/RISC-V contracts if the risk surface is accepted by the application.

MRC-721 and MRC-1155 provide native ownership, metadata, transfer, approval, and marketplace event shapes. Marketplace settlement can be native for the common case and contract-based for custom auction or game logic.

Why native does not mean larger attack surface

A common concern with native-module-heavy chains is that pushing primitives into consensus code grows the protocol's attack surface compared to a "thin L1 + many contracts" design. The opposite is true in practice for the workloads Monolythium serves.

A thin-L1 design spreads financial primitives across thousands of independently authored contracts, each of which can be wrong in its own way. A token standard reimplemented incorrectly in a single application contract has produced losses many times in EVM history. Activated native modules can centralize review for token accounting, NFT ownership, and released market behavior. Spending-policy enforcement remains an unactivated vNext module and has no current protocol-audit or composability claim. Applications may compose only against modules exposed by the pinned activated protocol version.

Native modules add to the consensus-critical line count. They subtract from the ecosystem-wide attack surface by far more. The chain pays the larger native module footprint to gain a much smaller composite surface across the applications that run on it.

16. Tokenomics

Design-status boundary. This section records the tokenomics charter and allocation design carried by the v5 paper. Unless a value is independently observable from finalized chain state, it must be read as a target parameter—not as proof of a current balance, activated reward path, bond requirement, vesting contract, sale term, yield, or enforceable holder right. The stale public RPC observed on 2026-07-16 is not suitable evidence for current economic operation. Agent-commerce fees, registry bonds, service-tier payments, delegation rewards, and the policies that depend on them remain separately activation-gated.

16.1 LYTH supply

The native token is **LYTH**.

- **Initial supply:** 100,000,000 LYTH at genesis.
- **Annual inflation cap:** 8% — a protocol-layer hard cap that applies to all cluster member and role rewards from the same allocation. The cap cannot be lifted by any signaling mechanism; lifting it requires a coordinated hard fork.
- **Treasury allocation:** pre-funded from the genesis reserve. There is no inflation tap for the treasury. The treasury draws on the genesis reserve and on buyback-burn yield over time.

The supply structure is constitutional: the inflation cap, the treasury’s funding mechanism, and the basic distribution rules are not changeable through any in-protocol process. They are part of the chain’s identity, not parameters open to runtime adjustment.

16.2 Allocation framework

The 100,000,000 LYTH initial supply is allocated across seven categories at genesis. Each category has its own release schedule and governance constraint; the supply itself is fixed at genesis.

#	CATEGORY	LYTH	SHARE
1	Community obligations (pre-allocated genesis balances)	32,781,503.90	32.78%
2	Foundation treasury reserve	15,000,000	15.00%
3	Ecosystem & grants	15,000,000	15.00%
4	Core contributors (vested)	13,000,000	13.00%
5	Public sale & community access programs	12,000,000	12.00%
6	Operator incentives & community programs	7,218,496.10	7.22%
7	Liquidity provision & integration support	5,000,000	5.00%
Total		100,000,000.00	100.00%

Category 1 — community obligations. A frozen ledger of 732 historical community-cohort entries, totaling 32,781,503.90 LYTH, is pre-allocated as on-chain genesis balances. No claim flow is required; each entry resolves to a direct balance on the holder's chain address. The ledger is canonical and immutable.

Category 2 — Foundation treasury reserve. Held in a multi-signature Foundation treasury with published signers and a published rationale requirement for every disbursement. The treasury does not draw on the 8% annual inflation; it operates from this category and from buyback-burn yield. Release is governed by treasury rules published by the Foundation.

Category 3 — ecosystem & grants. Reserved for developer grants, integration support, market-maker partnerships, infrastructure subsidies during the operator-bootstrap phase, and ecosystem programs that grow the chain's adoption surface. Released against published criteria over a multi-year horizon.

Category 4 — core contributors. Held in vested allocations to the engineering, design, research, and operations contributors who built the chain. Standard vesting includes a 12-month cliff and a 48-month linear vest. The published team and contributor list is the canonical record.

Category 5 — public sale & community access programs. The Genesis Liquidity Program and successor community-access offerings are funded from this category. Sale prices, vesting schedules, and KYC requirements are published at the time of each offering on the canonical project surfaces. Proceeds from these offerings, denominated in stablecoins and fiat, fund engineering payroll, infrastructure, market-maker arrangements, and external security audits across the consensus, signing, escrow, agent-commerce, interop-integration, and private-denomination surfaces.

Category 6 — operator incentives & community programs. Operator onboarding rewards, cluster-bootstrap incentives, community airdrops to historical ecosystem participants, and similar programs draw from this category. Programs are published at the time of activation.

Category 7 — liquidity provision & integration support. Reserved for external interop-provider integration and liquidity bootstrapping, market-maker arrangements on the native order book, and partnerships that deepen the chain's liquidity edge. Disbursements are public and reasoned.

16.3 Token utility and value capture

LYTH accrues utility across the entire chain surface, not only at the gas layer.

Consensus economics.

- **Operator self-bond.** Each active operator position requires 5,000 LYTH self-bond. At the target cluster topology of 700 active positions, this locks 3,500,000 LYTH as consensus-critical collateral. Equivocation slashing burns the entire bond; partial slashing for non-equivocation faults reduces it. The bond is the operator's economic skin in the game.
- **Standby capacity.** Each cluster's three standby operators also bond at the same level; at full capacity 300 standby positions × 5,000 LYTH = an additional 1,500,000 LYTH bonded.

Network economics.

- **Gas.** Every Monolythium transaction pays gas in LYTH — including transactions that move stablecoins, MRC assets, or NFTs. Stablecoin volume settled through the chain still requires LYTH to authorize the underlying transaction.
- **Fee burn.** A portion of every transaction's gas is burned, creating sustained deflationary pressure proportional to network activity.
- **Storage and state-growth fees.** State-heavy operations (large storage writes, persistent data) pay against state I/O metering, denominated in LYTH.

Service-tier economics.

- **Cluster service-tier payments.** The design denominates RPC and archival service contracts in LYTH and routes payment to the serving operator. This is not evidence of a current paid public endpoint. GPU prover requests and oracle-feed consumption remain gated; if armed, their exact market contract must be released and audited.
- **Interop fees.** Cross-chain interop runs through an external interop provider rather than an in-tree bridge, so the protocol no longer charges in-protocol bridge-route fees or runs bridge insurance and reserve programs. Value that arrives through the external provider still pays LYTH gas once it transacts on Monolythium.

Target registry economics (not activated for the §18 suite).

- **Discovery-registry bonds.** A future discovery registry may require an anti-spam bond under published terms.
- **Issuer-registry bonds.** A future issuer registry may use a separately specified anti-spam mechanism.

- **Arbiter-registry bonds.** A future arbiter registry requires published qualification, bond, and adjudication rules; no amount is quoted here.
- **Name-registry fees.** Each LYTH name registration pays a category-dependent fee under the U-curve pricing schedule. Renewal fees are recurring.

Target application economics.

- **Native order-book fees.** Maker and taker fees on the native CLOB pay in LYTH or in the asset being traded (with a LYTH-denominated discount path).
- **Spending-policy registration.** If the §18 protocol is activated, its versioned fee contract may require LYTH; no current fee is quoted here.
- **Escrow & dispute fees.** If activated, escrow, escalation, and arbiter compensation may settle under a released LYTH-denominated contract; this is not a current settlement claim.

Delegation economics.

- **Delegation tracking.** Delegators do not lock LYTH (liquid bonding), but their delegated balance is tracked against the per-wallet cap and earns reward distributions proportional to the cluster's performance.
- **Reward distribution.** The 8% inflation cap funds operator and delegator rewards. Service-tier revenue is direct to the providing operator; consensus rewards distribute through the cluster pool.

The target economic thesis is that an activated transaction consumes LYTH-denominated protocol resources regardless of the asset moved. If a future, separately released agent-payment integration settles an approved asset for an API call on Monolythium, that resulting on-chain transaction would require LYTH gas. No current USDC agent-payment route, hosted-MCP payment flow, interop settlement, or service-yield path is asserted here.

16.4 Canonical LYTH

The only canonical LYTH is the native token on Monolythium L1. Tokens with the same name on other chains — Binance Smart Chain, Ethereum, Solana, or any other — are not affiliated with the project unless explicitly listed on the canonical-tokens page published by the project.

Wallets, exchanges, and integrators should verify any cross-chain LYTH representation against the canonical-tokens page before treating it as official. Unaffiliated tokens carrying the LYTH name are not redeemable for native LYTH and do not carry the protocol-layer utility described above.

Bridged or wrapped LYTH on other networks, if and when such routes are established through an external interop provider, will be listed on the canonical-tokens page with the interop route, trust model, and verification status visible to users. Anything not listed there is not canonical.

16.5 Liquid bonding

Monolythium's staking model is **liquid bonding**: stakers retain custody of their LYTH at all times.

- Stakers never send their LYTH to a staking contract. Tokens remain in the user's wallet. The protocol tracks vote weight without locking the assets.
- There is **zero unbonding period** for delegators. A delegator can exit a delegation instantly. There is no slashing-window cooldown for delegators; slashing applies to operator self-bonds only.
- A user may multi-stake across up to ten clusters with basis-point granularity, subject to the per-cluster delegation cap below.

Liquid bonding gives delegators full flexibility while preserving the security model: operator self-bonds carry the slashing risk and are the principal capital at stake for protocol misbehaviour.

16.6 Operator self-bond and slashing

While delegators remain liquid, cluster operators guarantee network integrity with hard collateral:

- **5,000 LYTH self-bond** at onboarding into an active cluster position.
- **100% slash plus permanent operator exile on equivocation** (signing two distinct vertices at the same round). Recovery by re-bonding is not permitted; the operator identity is burned.
- **Reduced slashing for non-equivocation faults** (extended downtime, malformed-vertex broadcasting, cluster non-cooperation): partial bond burn plus temporary suspension.

The exemplary nature of equivocation slashing is deliberate. A protocol that tolerates double-signing invites it; a protocol that destroys the operator's stake and exiles them forever does not need to tolerate it.

16.7 Distributed delegation protocol

The core anti-capture mechanism is the **per-wallet delegation cap**: any single wallet can delegate at most X% of its total LYTH holdings to any single cluster. The cap binds on capital, not on wallet count.

The cap tightens progressively as the network matures:

NETWORK CONDITION	PER-CLUSTER CAP	MINIMUM DIVERSIFICATION
Early operating set	50%	2 clusters
Growing operating set	25%	4 clusters
Mature operating set	15%	7 clusters
Steady-state operating set	10%	10 clusters

Tightening is triggered by sustained network conditions — number of independent active clusters, geographic distribution, operator entropy — not by calendar dates and not by any in-protocol vote. Modifying the schedule requires a coordinated hard fork.

Wallet-multiplication resistance. The cap binds on capital per wallet, not on wallet count. An adversary splitting their capital across many wallets cannot route more LYTH to a single target cluster than they could from a single wallet — the per-wallet cap percentage applied to the smaller per-wallet balance produces the same nominal amount in aggregate. This eliminates the wallet-multiplication advantage that has historically broken delegated-stake systems.

The cap does not, and cannot, eliminate the underlying constraint that a sufficiently wealthy actor can route their capped percentage to any single cluster of their choice. That constraint is a function of capital, not of wallet count, and is unavoidable for any per-wallet cap. Three structural defenses combine to bound the practical concentration outcome:

- the per-wallet cap (this section),
- the service-weighted reward model (§16.8), under which delegated stake does not earn the staker a larger reward share — only proven service does;
- the per-operator multi-cluster cap (§6 and §17).

The combined effect is that even an actor with substantial capital cannot replicate single-validator dominance: their per-wallet cap limits their delegation to any one cluster, oversaturating that cluster with stake earns no extra reward share because the cluster pool is weighted by proven service rather than by stake (§16.8), and the per-operator cap prevents them from operating the cluster across many of its positions.

Stake division, not multiplication. A delegator with one hundred LYTH diversifying across ten clusters at the 10% cap delegates ten LYTH per cluster — total credited admission weight one hundred LYTH, not one thousand. Total credited weight is conserved. There is no economic gradient pushing voters toward cartel slates over independent diversification — cartel formation has cost but no incremental yield, and because the cluster reward pool is weighted by proven service rather than by stake (§16.8), routing more stake to a single cluster never increases the reward share that cluster earns.

16.8 Reward distribution

Inflation-funded rewards are distributed through **three independent streams**, none of which is weighted by stake:

1. **Base block reward** — an equal share to every active operator, on a round-robin schedule. Every active operator earns the same base reward regardless of its cluster's stake or service profile. This pays for showing up and participating in consensus.
2. **Proposer bonus** — paid to the round's proposer on the same round-robin rotation, so the bonus is shared evenly across operators over time rather than captured by any one cluster.
3. **Cluster pool** — sized within the 8% annual inflation cap and split across clusters by a **proved-service contribution score**, described below.

The cluster pool is service-weighted, not stake-weighted. A cluster's share of the cluster pool is determined by the **proven useful services it provides to the network** — not by how much stake is delegated to it. Each epoch, the protocol computes a per-cluster **service score** as the sum of proof-gated contribution terms:

- **base consensus participation** — a floor credited to every active cluster that participates in consensus;
- **archive service** — the heaviest single term, credited to a cluster that runs and serves full historical chain state;
- **GPU prover service** (gated off at genesis) — when armed, credited for proofs the cluster's operators generate and settle through the on-chain prover market;
- **public RPC service** — credited for serving wallet and application RPC traffic;
- **indexer service** — credited for serving indexed query traffic;
- **geographic and client diversity** — credited for distributing the cluster's operators across regions, networks, and implementations.

Every term is **earned only when proven**, and contributes **zero** when no proof is present. A cluster does not earn the archive term by declaring itself an archive node; it earns it by answering a reconstruction challenge from committed historical state. It does not earn the prover term by claiming GPU capacity; it earns it by settling verified proofs on-chain. Stake is never an input to this score.

Stake is an admission gate, not a reward weight. A cluster's stake — operator self-bond plus delegated stake — does exactly one thing: it sets the cluster's **rank for the top-100 active-set selection** (§17). Once a cluster is in the active set, its stake no longer influences anything. Active clusters vote with **equal weight** (consensus quorum is by count, with a round-robin leader, §12), and they earn from the cluster pool strictly in proportion to their **proved service**, not their stake. Two clusters with identical stake but different proven service earn different shares; two clusters with identical proven service but different stake earn the same share.

This makes decentralization the highest-yield strategy on the contribution axis rather than the capital axis: the way to earn more is to **serve more** — run the archive, generate proofs, serve RPC and indexer traffic, diversify across regions — not to accumulate delegated stake.

Honesty about proofs. The service terms differ in how trustlessly they are proven, and this is stated plainly rather than overclaimed:

- **prover** is **cryptographically proven** — the proof either verifies on-chain or it does not;
- **archive** is proven by a **reconstruction challenge** — the cluster must produce historical data that hashes to a root it committed at registration, or it earns zero for the term and is liable to slashing;
- **RPC, indexer, and diversity** are, in the first release, **attested and bonded**: a rotatable foundation probe authority attests reachability on-chain, diversity inputs are self-declared but bonded with an anti-multiplicity cap, and no purely self-declared claim mints reward. These attested terms harden toward **trustless challenge-response** over time; the first release does not claim they are already trustless.

Constitutional framing. The service-contribution model — that the cluster pool is split by **proven service, not by stake** — is the **design commitment of the protocol**. The per-service weights and the proof parameters (which services count, the floor and caps, the archive challenge frequency, the attestation window) are the **calibration**, adjustable **only through hard-fork protocol updates**. There is no on-chain governance to retune them. The commitment is constitutional precisely because, if the weighting rule were vote-tunable, a large capital holder could vote to reintroduce stake-weighting and recapture the reward layer; putting both the rule and its calibration behind a hard fork keeps it constitutional.

Within a cluster, the consensus reward pool distributes equally among its operators. Service-tier revenue (RPC, archive, prover, oracle) is per-operator direct rather than pool-split; an operator with a strong service offering captures the direct revenue from that service. Delegators continue to earn **pro-rata of their own delegated stake** from the cluster's delegator share (the charter below), so a delegator's return tracks the stake they personally contributed even though stake does not weight the cluster's pool share.

Cluster charters. The intra-cluster split is a default, not a constant. A cluster formed without a charter distributes its operator-side pool equally among its members and routes 50% of its cluster reward pot to its delegators. At formation, the members may instead bind a **charter** into the signed formation consent: a per-member share table for the operator-side pool, the cluster's delegator share — bounded below by a 20% protocol floor — and a consent expiry. The charter is part of the consent digest every member signs, so each operator signs the exact economic terms of the cluster being formed; no member can be bound to terms they did not sign. Clusters formed without a charter retain the defaults above.

16.9 Auto-rebalance on cap tightening

When a tightening transition activates and the cap moves (for example from 25% to 15%), existing over-cap delegations are not slashed or instantly invalidated. The protocol enters a **grace period**:

- over-cap delegators receive standard rewards on the in-cap portion;
- over-cap delegators receive a tapered share on the over-cap excess, sized to give honest delegators sufficient time to rebalance without meaningful yield loss;
- the wallet UX (autovote routines, see §25) is notified to rebalance;
- after the grace period ends, the over-cap excess earns no rewards until the delegator reduces per-cluster allocation or redirects the excess to additional clusters.

The transition is non-disruptive: delegators have a clear economic signal and a defined window to adjust; clusters have continuity of voting weight during the rebalance.

16.10 Acknowledged limitations

Two limitations are documented honestly.

Custodian aggregation. A large custodian holds customer assets in a pooled omnibus wallet. The single wallet is subject to the cap, potentially under-representing the diverse preferences of underlying users. The cap still binds on the custodian's pooled wallet itself — even a major exchange under-representing many users cannot dominate any single cluster beyond the cap. The protocol does not solve sub-account expression of preference, but it prevents custodian-level cluster capture. Sub-account delegation standards and integration with proof-of-personhood systems are areas of ongoing development.

Correlated preferences. The protocol does not, and should not, prevent legitimate market behaviour. If a large majority of independent, sovereign delegators happen to choose the same high-performing cluster, that cluster legitimately earns a large share of delegated stake. This is a reflection of the market's will, not a Sybil attack.

17. Cluster Operations: DVT, Slashing, Service Tiers

17.1 Distributed validator technology

A cluster is a **distributed validator** in the strict sense: ten independent operators jointly produce one logical cluster vertex per round, each signing the cluster's vertex with its **own independent ML-DSA-65 key** — a key only that operator holds and controls. Every operator signs with the key it alone controls; nothing is held jointly across the cluster. A vertex is valid once a 7-of-10 quorum of those independent operator signatures is present, expressed as a **bitmap multisig**: an operator bitmap recording which operators signed (bit *i* corresponds to the cluster's operator *i*), followed by their raw individual signatures. It is not a combined threshold signature and requires no aggregation ceremony to produce — any seven of the cluster's ten members' individual signatures form a valid certificate. The cluster signs one vertex per round. Because only seven of the ten members' signatures are needed, up to three operators can be offline without halting the cluster's participation.

Distributed validator technology gives the chain three properties single-operator validators cannot:

- **Operator-level fault tolerance.** A single operator's outage does not stop the cluster.
- **Operator diversity.** Operators within a cluster can be in different regions, on different network providers, on different hardware classes, and under different jurisdictions.
- **Operator markets.** Operator capacity is a public market — operators publish capabilities, clusters compose themselves from the market, and reputation accrues to the operator identity over time.

17.2 Standby capacity

Each cluster has a **standby of three operators** in addition to its seven active members. Standbys participate in the cluster's communications, are known to the cluster roster with their own independent ML-DSA-65 keys, and are ready to step in when an active operator goes offline, is rotated out, or departs.

This is what makes the cluster operationally resilient, not just structurally so. Because a valid certificate needs only seven of the cluster's ten members' individual signatures, **failover is automatic and free**: when an active operator goes offline, the signatures of the remaining online members — a standby's included — fill the 7-of-10 threshold on their own. There is **no promotion ceremony and no key handover**; each member already signs with the independent key it has always held, the 7-of-10 quorum continues to hold over whichever seven members are online, and the cluster keeps producing.

17.3 Service tiers

The following are **target economic service tiers**, not evidence of current user-paid revenue or an activated public fee API:

- **GPU proving** (gated off at genesis). If armed, off-chain proof generation could run on GPU-equipped operators with on-chain verification on commodity-CPU operators under a released market contract. The prover market is **not reachable in the public v0.4.0 protocol**.
- **RPC and archival**. The target model lets clusters serve wallet RPC calls, indexer queries, and archival reads under a released LYTH-denominated service contract.
- **Oracle feeds** (service-tier oracle-feed payments gated off at genesis). Clusters with reliable infrastructure can participate in the oracle aggregation network and earn LYTH from oracle consumers once the service-tier payment path is armed.

The design routes service-tier revenue **directly to the operator** providing the service rather than through the cluster reward pool. Exact activation, prices, proofs, and payouts require versioned protocol evidence.

17.4 The GPU service tier separation

A cluster's GPU prover capacity is a **service tier**, not a consensus dependency. A cluster's consensus participation depends on its 7-of-10 ML-DSA-65 bitmap multisig over CPU-bound operators; GPU is required only for off-chain proof generation. A GPU operator going offline costs the cluster its prover service revenue but **does not affect the cluster's consensus participation or the chain's Byzantine fault threshold**.

The GPU-as-service-tier choice is deliberate. Enforcing GPU on every operator would create hardware-cost moats that smaller operators cannot cross, producing operator monopolies and hardware-class centralisation. The chain's design lets operators specialize: clusters that want strong GPU service hire GPU-equipped operators; clusters that don't earn the consensus base reward and avoid the GPU capital cost. The market prices the difference; the consensus layer remains commodity-CPU and accessible.

The asymmetric cost economics make this work. Proof generation on a modern GPU runs in tens of seconds; on-chain verification by commodity CPU operators runs in a small, predictable gas budget. Generation is expensive and concentrated; verification is cheap and distributed.

18. Target Agent-Commerce Architecture

This section is a **design specification for a future versioned activation**. The current public development network does not expose these eight primitives as an executable suite, and the names, calls, state machines, events, fees, and layer assignments below are not a current ABI. Each primitive requires a separately versioned protocol implementation, SDK contract, wallet rendering, privacy review, migration plan, and release gate before an application may rely on it.

#	PRIMITIVE	TARGET LAYER	INTENDED PURPOSE
1	attestation	Target native module	Signed-hash + typed-schema registry
2	consent	Target native module	Principal-signed consent records with revocation
3	issuer-registry	Target native module	Permissionless registry of credential issuers
4	discovery	Target native module	Permissionless service-listing registry
5	reputation	Target indexer view	Multi-dimensional aggregation over finalized events
6	availability	Target native module	Availability state composed by domain registries
7	escrow + arbiter-registry	Target native modules	Escrow, counter-offer, and pluggable arbitration
8	spending-policy	Target consensus-critical module	Per-account spending constraints enforced at admission

In this design, only `spending-policy` would need the consensus path, because a delegated agent must not be able to bypass its principal's limits. Reputation would remain an indexer view because Sybil ratings, wash trades, and coordinated reputation farming do not belong in consensus-critical code. The remainder of §18 uses present tense to state desired invariants compactly; it does **not** indicate deployed code.

18.1 Attestation target

The foundational primitive. A `subject` is attested by a `signer` against a `typed schema`; the resulting attestation is content-addressable, revocable by the signer, and verifiable offline against the signer's public key. New schemas register at runtime — no hard fork is required to add a new attestation type.

Use cases:

- capability claims (“this entity is qualified to do X”);
- consent attestations (principal signs the terms);
- reputation events (counterparty rates an interaction);
- service deliverables (provider attests to completion).

18.2 Consent target

The target consent registry would enable principal-signed records with controlled revocation. The following calls are illustrative pseudocode, not callable endpoints:

- `grant(scope, terms, expiry?) -> ConsentId`
- `revoke(id)` — instant for new uses; existing in-flight escrows initiated under the consent are grandfathered until completion.
- `query(scope) -> Vec<Consent>` — read all currently-active consents for a scope.

Revocation is instant for new activity; in-flight workflows are grandfathered until completion. This balances the principal’s right to instant revocation with the counterparty’s right not to be left holding a half-completed work product because the principal changed their mind mid-stream.

18.3 Issuer-registry target

The target is a permissionless map from “issuer name + jurisdiction” to “public key + metadata” for credential-issuing authorities, with an anti-spam mechanism whose exact asset and amount are still release-gated. The protocol would not pick winners; verifiers would decide which issuers they trust.

Use case: when a holder presents an attestation signed by an issuer (a state bar, a medical board, a credentialing program), the chain knows which public key corresponds to that issuer’s name. A verifier offline-verifies the attestation against the registered key without trusting any single party’s claim of who the issuer is.

The protocol has no mechanism to remove an issuer from the registry once registered. If an issuer turns out to be fraudulent, the market response — verifiers stop trusting the issuer’s signatures, attestations under that issuer’s key lose value — is the only mechanism. Attempting to add a removal mechanism would be governance-shaped and is excluded by the chain’s no-governance design.

18.4 Discovery-registry target

The target is a chain-native, permissionless service-provider registry. Its listing tuple and any anti-spam bond below are schema requirements, not deployed fields or a quoted fee. A future chain release could emit indexable events while off-chain indexers build the user-facing search experience.

Listings are organized under a hybrid taxonomy:

- **fixed top-level categories** for filtering (legal, dev, design, finance, consulting, marketing, social, operations, and others);
- **free-text description fields** per listing;
- **embeddings** for fuzzy and natural-language search, computed off-chain by indexers from listing text; the on-chain storage is the listing itself.

Marketplace scope is universal. The same discovery primitive serves lawyers, developers, designers, finance professionals, AI agents, paid one-off consultations, batch promotional services, and any other category that fits a typed listing schema. Same registry, same primitive, different category.

The stake bond is sized to deter spam, not to gate participation; it is held for the lifetime of the listing and returned on closure. Frivolous spam listings cost LYTH; legitimate providers pay a one-time entry cost.

18.5 Reputation target

Reputation is **derived from on-chain history** — there is no separate “reputation token” or “rating contract.” The reputation views aggregate over the same on-chain data that backs every other primitive.

The chain emits typed events for every reputation-relevant interaction: escrow released, deliverable attested, dispute resolved, counter-offer countered. Off-chain indexers aggregate the events for fast queries; raw data is provable from the chain.

Reputation is **multi-dimensional** — four fixed axes: speed, quality, communication, accuracy. Each axis is rated 1-5 by counterparties; aggregations are public. The four axes give downstream search clients enough signal to filter on dimension (“fast turnaround” vs. “high quality” vs. “good communication”) without forcing a one-dimensional collapse.

Reputation is **portable**. An agent’s reputation address is the agent’s chain address. Reputation accrues to the agent address. If the agent switches model provider, the address stays the same; the reputation persists. This is the structural-portability property that closed payment systems cannot offer.

18.6 Availability target

A generic availability state machine composed by every domain registry — arbiters, providers, inference services, all of them. One source of truth for:

- **open request count** (how many requests are currently outstanding for the entity);
- **accept/deny per request with timeout** (a request is auto-declined if not accepted within a configurable timeout);
- **vacation/availability flag** (“not taking requests until date X”).

The same state machine serves an arbiter (tracking how many disputes they have queued), a provider (tracking how many open offers they have outstanding), and an inference service (tracking how many concurrent jobs are being executed). One implementation, one audit, many consumers.

18.7 Escrow-and-arbiter target

The target uses two versioned modules — provisionally called `escrow` and `arbiter-registry` — to compose a counter-offer and dispute-resolution flow. Neither module is activated on the current public development network.

The arbiter mode is set per escrow at creation:

- **Single arbiter** — one designated arbiter signs to release or refund. Suitable for low-value escrow.
- **Quorum arbiter** — N-of-M signatures from a designated arbiter set. Suitable for mid-value escrow.
- **Human-required** — arbiter must be a human-credentialed entity per attestation. Suitable for high-value escrow.

Arbiters would be **pluggable** through a permissionless registry. Bootstrap policy, qualifications, anti-spam terms, conflicts, appeals, and liability must be published with the eventual protocol; this paper does not assign the Foundation an undisclosed arbiter set or special production authority.

The counter-offer flow is a full on-chain negotiation state machine:

```
Open → Negotiating(round_n) → Accepted → InProgress → Submitted →
Released | Disputed
```

- counter-offers can modify any term (timeline, scope, price, payment schedule, penalty clauses);
- each counter-offer is signed by both parties before becoming binding;
- either party can walk away during negotiation with no penalty — escrow funds are not yet locked until `Accepted` ;

- a practical round limit applies before the offer auto-closes (anti-griefing).

Reviews and disputes are distinct: reviews are ex-ante quality signal handled by reputation; disputes are ex-post fund recovery on a specific transaction, handled by escrow. They do not fuse.

18.8 Spending-policy target

This is the only target agent-commerce primitive proposed for the consensus path. If activated, the protocol must guarantee that a transaction signed by an agent account violates no registered policy constraint, because the delegated agent cannot be trusted to enforce its own caps.

The model is **sub-account**. Agents are sub-accounts of a human or organizational principal. The principal creates a sub-account for the agent, assigns LYTH to the sub-account, and registers a spending policy on the sub-account. The agent can sign transactions from the sub-account, but the protocol enforces:

- per-transaction caps;
- per-day, per-week, per-month rolling budget caps;
- per-counterparty allow-lists;
- per-category allow-lists (legal, dev, design, etc.);
- time-of-day windows;
- expiry dates;
- revocation by the principal at any time.

The design requires content-addressed policies in consensus state and admission-time rejection of violations. Those are target invariants, not behavior available through the current public RPC.

18.9 Economic-runbook target

The target requires agents to select **typed, versioned, wallet-visible action templates** rather than improvise free-text RPC. The following names are a design vocabulary, not registered methods or current MCP tools:

- `pay_vendor`
- `open_escrow`
- `release_escrow`
- `refund_payment`
- `book_service`
- `place_trade`
- `set_spending_policy`
- `revoke_agent_permission`

- `verify_receipt`
- `rate_vendor`

The intended safety property is a bridge from natural language to a constrained, human-authorized workflow. It depends on the wallet and activated protocol validating the same versioned schema. Until that contract exists, an application must not present any name above as executable.

18.10 Stele product boundary

Stele's release architecture is a standalone public web product, not a marketplace embedded in the desktop wallet. As of 2026-07-16, the public preview is live at stele.monolythium.com, and its catalog contains zero published services. Browser Wallet v0.4.5 is a prerelease, while Stele's hosted services never receive a seed phrase or private key. The public web authenticates users through Browser Wallet. That authentication proves only current control of the selected wallet address; it does not prove a human or legal identity, durable ownership, credentials, or authority. Its wallet-authenticated Provider Studio at stele.monolythium.com/studio can create, edit, preview, and delete private wallet-owned provider-listing drafts. These durable provider-listing drafts are not published, discoverable, or transactable, and provider publication remains off. Booking-approval drafts are separate: the web can inspect an existing valid non-economic booking-approval draft, but it does not create booking-approval drafts. Inspection is read-only and non-economic: the public web exposes no booking, payment, settlement, or other economic controls.

SURFACE	PUBLIC RELEASE BOUNDARY	SEPARATELY GATED TARGET
Public web	Live public preview; browse the empty public catalog; authenticate through Browser Wallet with proof limited to current control of the selected address; use Provider Studio at stele.monolythium.com/studio to create, edit, preview, and delete private wallet-owned provider-listing drafts; inspect an existing valid non-economic booking-approval draft; the web does not create booking-approval drafts and exposes no booking, payment, settlement, or other economic controls	Provider publication, private rooms, settlement, disputes, and reviews
Hosted Stele MCP	Keyless and OAuth-protected; authenticated <code>tools/list</code> is the authoritative inventory; baseline public catalog search and bounded non-economic booking-draft preparation; hosted booking-draft preparation is unavailable without a published listing; <code>stele_create_provider_listing_draft</code> exists only when that exact name appears in authenticated <code>tools/list</code> ; when listed, each successful operation creates exactly one new private wallet-owned unpublished provider-listing draft, while an idempotent replay returns the same draft without creating another; no existing-draft list, read, update, delete, or publish operation; no signing, submission, broadcast, or settlement	Any future economic or transaction capability requires a separately reviewed release and explicit user authority
Isolated local Stele MCP	Exactly three read/status tools in the current package	Any transaction tool requires a separately reviewed local signer/SDK contract and explicit user authorization
Browser wallet	v0.4.5 prerelease: prove current control of the selected wallet address and retain human authority; authentication does not prove human or legal identity, durable ownership, credentials, or authority	Display and sign a released, versioned economic transaction only after protocol activation
Desktop wallet	Target product line retires the Stele marketplace surface; generic wallet functions remain independent	Legacy or unreconciled desktop builds may retain a gated historical

SURFACE PUBLIC RELEASE BOUNDARY		SEPARATELY GATED TARGET
		surface until reviewed removal and migration ship

The two draft classes are independent. A provider-listing draft is a durable server-side, wallet-owned candidate service record created in Provider Studio; it can exist while the catalog is empty and has no publication or transaction action. Here, private means access-controlled to the authenticated wallet and absent from the public catalog, not local-only storage or end-to-end encryption. A booking-approval draft is a separate, short-lived, non-economic review record prepared through hosted MCP for an existing published listing. Provider Studio does not create it. For hosted MCP, authenticated `tools/list` is authoritative: `stele_create_provider_listing_draft` exists only when that exact name appears there. When listed, each successful `stele_create_provider_listing_draft` operation creates exactly one new private wallet-owned unpublished provider-listing draft; an idempotent replay returns the same draft without creating another. The tool cannot list, read, update, delete, or publish an existing provider-listing draft. Neither draft review nor creation can make a transaction.

The live deployment establishes only public-preview and private-drafting availability: zero published services means it does not evidence a provider market or on-chain listing registry. Hosted booking-draft preparation is unavailable without a published listing; with zero published services, no valid booking-approval draft can currently be prepared. Catalog records, provider-listing drafts, and booking-approval drafts do not prove on-chain listings, availability, price, escrow, reputation, or settlement. E2EE rooms, provider publication, economic writes, transaction signing, and mainnet remain off until their independent release gates are satisfied. No hosted component may hold wallet secrets or convert either draft class into a transaction.

19. Privacy Cordon

The bifurcated denomination (§5) is enforced by a native **caller-origin cordon** in the execution layer.

The cordon is a single rule, stated simply: **a public contract or module path cannot receive private-denominated value, and private-denominated state cannot be used in any public contract context.**

Concretely, the runtime tracks the denomination origin of every value passing through a contract call frame. A call that mixes denominations is rejected at admission. A contract that attempts to read private-denominated state from a public-denominated execution context is rejected. The rule is enforced by the host, not by user code, so no contract can opt out.

This is the rule that turns the bifurcation from a policy into a structural property. Without the cordon, a sufficiently clever contract could route private LYTH through a public contract and emit it on the public side. With the cordon, that path does not type-check.

Privacy modes inside the private denomination

Some of these are live today; the confidential (amount-hidden) value path is **not present in the protocol**. Live now:

- **stealth addresses** for sender/recipient unlinkability;
- **per-account and per-asset privacy policy** — which privacy levels are legal for an asset and whether KYC is required.

Removed from the protocol — the confidential value path is not implemented and is not reachable at any height. It is described here as a future capability, not a gated one. The earlier design (a Ristretto/Pedersen-commitment construction with Bulletproofs range proofs) was **deleted**, not merely disabled: an audit found the commitment path could mint unbacked value, and a disabled precompile is a weak guarantee — under a signed upgrade channel a single configuration entry could re-arm it. Removing the construction removes that possibility. Any future confidential-amount capability will be a new, post-quantum construction introduced deliberately:

- **confidential transactions** for amount hiding;
- the **one-way crossing** from public to private (public→private), with no reverse path;
- **transfer to another private (amount-hidden) address**;
- **burn** from the private denomination.

In practice today, you can send to a stealth address and set privacy policy, but you cannot move confidential (amount-hidden) value on-chain — the protocol has no code that would do so. The description of the private denomination in this section and in §4.4 and §5 is the intended design for when that path is armed.

A full shielded pool inside the private denomination is not part of the base protocol. The structural separation between public and private is the novel primitive; layering additional privacy machinery on top of bifurcation before bifurcation itself is operationally proven would be premature.

Privacy at the wallet edge

Wallets render the two denominations distinctly. A user moving LYTH from public to private sees the warning explicitly: “this is a one-way move; private LYTH cannot return to public.” The crossing is intentional and visible, not an accident.

20. Interop and the Liquidity Edge

Monolythium needs liquidity but does not need to inherit EVM execution to get it — and it no longer ships an **in-tree bridge** to get it either. The entire in-tree bridge stack — the on-chain bridge proof verifier, the route-policy, fee, and insurance scaffolding, and the associated build feature — has been **removed** from the protocol. The chain does not verify bridge proofs on-chain.

Cross-chain interoperability is instead delivered by integrating an **external interop provider** (evaluation in progress; described here vendor-neutrally). The rationale is deliberate: running the chain's own bridge verifier in-tree concentrated the highest-risk, audit-heavy surface inside consensus. Moving interop to an external provider takes that surface out of the core, lets interop evolve without a re-genesis, and keeps the base layer post-quantum and lean.

The liquidity strategy is now two layers:

1. **External interop-provider integration** for moving major external assets in and out. Verification, routing, and settlement of the cross-chain leg are the provider's responsibility, not the protocol's.
2. **Issuer-supported native assets** where the network earns enough adoption to justify direct issuer integrations.

Wrapped assets are labeled honestly. An asset that arrives through the external interop provider is not the same as a native issuer-minted asset. Wallets and explorers surface the provider, route, trust model, and any provider-side risk metadata. **The goal is not to hide interop risk; the goal is to make interop risk legible.**

20.1 Interop risk is legible at the wallet edge

Because the chain does not itself verify the cross-chain leg, the trust model for any incoming bridged asset is the **external provider's** trust model, and the wallet surfaces it rather than implying the protocol has verified it. Users often treat all bridged assets as equal. They are not equal: an asset that arrived through a trusted-multisig provider, an asset from a light-client- or proof-verified provider route, and a native issuer-minted asset all carry different risks. The wallet and explorer ecosystem makes those differences visible before a user signs, and marks clearly which assets are native and which arrived through an external provider.

20.2 What the chain no longer does

For the avoidance of doubt, the following were part of earlier drafts of this section and are **no longer part of the protocol**:

- there is no in-tree light-client or zero-knowledge bridge verifier, and no on-chain bridge-proof verification;

- there are no protocol-level bridge route-policies, drain caps, circuit breakers, per-route cooldowns, or bridge insurance/reserve programs;
- there is no SP1-Helios or CCIP verifier stack and no `sp1-bridge-verifier` build feature;
- the `0x1008` bridge precompile slot is retired to a fail-closed tombstone that rejects any call, and the node refuses to boot any milestone that would attempt to activate it at any height.

Any drain-cap, circuit-breaker, cooldown, or route-verification behavior that an incoming bridged asset carries is a property of the **external interop provider**, surfaced to the user as provider metadata — not a guarantee enforced by Monolythium consensus.

21. Hardware Sovereignty

Production operators run on **Monarch OS** — an immutable substrate built on a minimal Linux base, hardened for institutional security and designed for permissionless accessibility. A home operator with a recent gaming PC runs on the same substrate a co-located bare-metal operator runs.

21.1 Substrate properties

- **Immutable image.** Monarch OS is delivered as a single signed image. There is no package manager, no SSH, no interactive shell. The system is not a general-purpose Linux distribution; it is a purpose-built operator substrate.
- **Verified rooted filesystem.** Every block of the filesystem is verified against a signed Merkle root at read time. Tampering with the filesystem is mathematically impossible without breaking the boot.
- **TPM 2.0 measured boot.** The boot sequence's hash measurements are extended into the TPM's Platform Configuration Registers. The system can prove what it booted via TPM PCR quotes.
- **TPM-sealed operator signing key.** An operator's ML-DSA-65 consensus signing key is sealed against the local TPM. Each operator holds its own independent key — there is no cluster-wide key and nothing held jointly, so a broken seal affects only that one operator. If the chassis is opened or the firmware is modified, the seal breaks and the operator can no longer sign with that key until they re-onboard with a fresh key; the cluster keeps producing on the remaining members' signatures because only seven of its ten members are needed.
- **No transitive dependency surface.** A conventional Linux distribution inherits a vast supply-chain attack surface through transitive dependencies. Monarch OS has no transitive-dep mechanism. An operator never installs a transitive dependency, so it cannot inherit one.
- **No userspace foothold for kernel exploits.** A modern class of kernel local-privilege-escalation bugs requires the attacker to first execute code as a local non-privileged user. Monarch OS structurally denies that foothold: no SSH, no interactive shell, no multi-user system, no package manager, no cron, no general-purpose userspace daemons. The only userspace process of consequence is the node binary itself.

21.2 Kernel attack-surface hardening

Beyond the absence of a userspace foothold, Monarch OS ships with an aggressively trimmed kernel configuration: features the operator node does not need are not compiled in or are compiled out of autoload.

- The userspace cryptographic kernel API is disabled at kernel-config level. The node binary performs all cryptographic work in-process via Rust libraries (ML-DSA-65, ML-

KEM, BLAKE3), not via kernel crypto sockets. The entire class of kernel-CVE attacks against userspace crypto APIs is closed off going forward.

- **Unused kernel subsystems are disabled:** virtualization (operators do not host VMs), audio and video, Bluetooth, wireless drivers (operators are wired-network-only), legacy filesystems, and other categories irrelevant to a server-grade operator workload. The smaller the kernel surface, the smaller the future-CVE exposure.

The crypto-on-OS / crypto-in-app distinction is structural. A chain that signs blocks via system-level cryptographic invocations has a fundamentally different exposure profile than one that signs blocks via in-process Rust libraries. Monolythium chose the latter from genesis. Kernel-CVE classes that target userspace crypto APIs do not affect the operator's signing path at all.

21.3 Hardware targets

Monarch OS runs on standard x86-64 hardware. Three target categories:

- **Home server.** Dedicated tower or rack server. Recent professional or consumer CPUs, 32–64 GB RAM, NVMe storage. Firmware TPM available on Intel/AMD CPUs since the late 2010s. Suitable for a home operator running a single cluster position.
- **Gaming PC.** Consumer hardware with firmware TPM. Suitable for a hobbyist operator. Performance is typically bounded by network and storage, not CPU.
- **Bare-metal co-location.** Dedicated server in a co-location facility with discrete TPM 2.0 chip. Recommended for operators offering high-bandwidth service tiers (RPC, prover, oracle).

GPU is **optional** for consensus and required for the GPU prover service tier. Operators who want to earn from the prover market install consumer or workstation-class GPUs; operators who do not, do not. Consensus participation is unaffected.

21.4 Continuous runtime attestation

Operators publish **TPM PCR quotes** every epoch to the on-chain node registry. The PCR quote is a signed declaration of the values currently in the TPM's PCRs at the time of quote generation; it proves what the operator's machine is running.

The `pcr_values` field captures the state of measurement: the bootloader, the kernel image, the substrate root filesystem, the running node binary. A change in any measured component changes the corresponding PCR. The chain stores the attestation; explorers display it; cluster members and other operators verify against expected PCR values.

A change in PCR values that is not preceded by a coordinated upgrade announcement is treated as an anomaly. Cluster members are notified; the cluster's quorum can vote to suspend the operator pending investigation.

This makes it cryptographically observable when a corporate hypervisor or unauthorized management layer is introduced underneath a Monarch node. The PCR values would shift; the network would notice; the cluster would respond.

21.5 Network and geographic diversity

Hardware attestation alone does not prevent a single entity from running multiple registered operators on physically distinct hardware in the same datacenter, on the same upstream network provider, or in the same jurisdiction. The protocol augments per-operator hardware attestation with **network-level and geographic diversity scoring** at cluster admission and during continuous attestation:

- distinct IP address per operator is a baseline requirement;
- ASN (Autonomous System Number) is recorded on-chain and entered into cluster admission scoring;
- geographic region claims are cross-checked against IP geolocation and latency triangulation from the existing cluster set;
- hosting class (bare-metal, co-location, cloud) is derived from PCR attestation patterns and network classification.

A cluster's diversity score is visible to delegators. Diverse clusters earn delegator preference; concentrated clusters lose it.

22. Threat Model

Monolythium's threat model is adversarial across the chain's full lifecycle. The chain assumes:

- some operators are Byzantine;
- some clusters may be compromised in part;
- external interop is an active attack surface;
- private keys can be stolen;
- model providers can be compromised;
- governance can be socially engineered (which is one reason the chain has none);
- cryptographic primitives may eventually fail;
- nation-state-scale attackers can target the protocol and its dependencies.

The chain's defensive posture is **separation of blast radius**. Different surfaces have different protection budgets, and a failure at one surface should not compromise another.

22.1 Surface separation

SURFACE	PRIMARY PROTECTION	FAILURE SCOPE
Consensus	Cluster threshold + equivocation slash + DAG-BFT mathematics	A failure here halts safety; the protocol must reject equivocating operators and degrade gracefully
Cryptography (user signatures)	ML-DSA-65 (only); a break-day migration is handled by process — emergency freeze (§23.2) plus a coordinated hard fork	A primitive break is a coordinated migration, not an armed on-chain rotation; the in-protocol emergency-key registry and algorithm rotation were removed and return in a future genesis (reconciliation note item 10)
Interop (external provider)	Cross-chain verification and routing run with an external interop provider, outside consensus; wallets surface the provider's trust model and risk metadata	An interop failure is bounded to the external provider and the assets that transited it; Monolythium consensus and native accounts are unaffected, because the chain neither verifies nor custodies the cross-chain leg
Mempool	Plaintext mempool; transaction-ordering fairness handled at the DAG-consensus layer	Transactions are visible before inclusion; ordering follows deterministic DAG linearization rather than a single sequencer's discretion
Application contracts	Audit + native modules + sandbox boundaries	A contract bug damages the contract's users; native modules and the consensus layer are insulated
Hardware	TPM PCR attestation + immutable substrate + network/geographic diversity scoring	A compromised operator is detectable through PCR drift; a compromised hosting class is detectable through diversity scoring
Recovery	Emergency freeze (global pause) + coordinated hard fork	The in-protocol emergency-key registry / frozen-account rotation was removed ; public operational playbooks are a mainnet gate, not an on-chain recovery ABI

The point is that the chain does not have a single point at which “everything depends on this.” Each surface has its own defense, and each defense has bounded failure consequences.

22.2 What the chain does not promise

The chain does not promise:

- that operators will not be compromised individually — instead, the cluster threshold and DVT structure bound the damage;
- that an external interop provider will never be exploited — instead, keeping interop outside consensus bounds the damage to the provider and the assets that transited it, and wallets surface the provider's trust model before a user relies on it;
- that smart contracts will not have bugs — instead, native modules carry the audited primitives and contracts are sandboxed;
- that cryptography will work forever — instead, a future ML-DSA-65 break is handled by a coordinated migration (the emergency freeze in §23.2 plus a hard fork); the earlier in-protocol emergency-key registry and algorithm rotation were removed and are a future-genesis capability, not a present-tense one;
- that no user will lose funds to social engineering — a future released wallet and activated runbook/spending-policy model would only bound delegated authority; neither is a current Steele or hosted-MCP guarantee.

Honesty about limits is part of the threat model. A chain that claims invulnerability is a chain whose users are unprepared when invulnerability fails.

22.3 What about MEV?

MEV — value extracted by privileged actors who can reorder, insert, or censor transactions — is bounded structurally by the chain's design.

- **Plaintext mempool.** The mempool is plaintext, so a transaction body — including a CLOB order — is visible before inclusion. Mempool-visibility front-running is therefore not foreclosed at the mempool layer; it is bounded instead by deterministic DAG ordering and the native order book, below.
- **No leader seed to grind.** Wave leadership is a deterministic round-robin schedule over the active cluster set (§11), not a seed-derived lottery. Because there is no aggregate seed, there is nothing for an adversary to bias by withholding or selectively releasing a partial signature — the leader-grinding class is foreclosed by construction. The trade-off is stated plainly: leadership is fully **public and predictable**, so a scheduled leader knows in advance that it will propose. That is a deliberate property of the pacemaker schedule, not a claim of unforgeability; transaction ordering within a leader's proposal is a separate consideration that predictable leadership does not by itself foreclose.
- **Native order book.** The native CLOB settles orders deterministically. Sequencer-style MEV games against an order book are bounded by the consensus order, which

is determined by the DAG linearization rather than by a single sequencer's discretion.

MEV is not zero, and the chain does not claim it is. Some MEV (backrunning of public events, arbitrage between markets, latency-based capture) exists wherever transactions are visible, and the mempool is plaintext. The chain's design forecloses the leader-grinding and sequencer-discretion categories through the seedless round-robin leader schedule and deterministic DAG ordering, and leaves the remainder as a competitive market that benefits ordinary users through tight spreads.

23. Recovery and Emergency Posture

The chain's recovery posture is designed for two scenarios that are normally underspecified in L1 protocols: a cryptographic primitive breaks, and a coordinated systemic event requires the chain to freeze in place while it is sorted out.

23.1 Crypto agility — deferred to a future genesis

REMOVED — see the top-of-file reconciliation note (item 10). The in-protocol **emergency-key registry** and the SLH-DSA (FIPS 205) hash-based emergency-backup key described below have been **removed** from the protocol. An audit found the advertised break-glass inoperative — the registered backup could not actually be used to rotate — so rather than ship a non-functional recovery primitive, the chain is **ML-DSA-65-only** at launch with no on-chain emergency algorithm rotation. Post-quantum crypto agility (a cross-family backup primitive plus an in-protocol rotation path) is intended to return in a **subsequent genesis**. The paragraphs below describe that future design, not a live feature.

The intended design was for a user to pre-register a backup key in a different cryptographic family (specifically SLH-DSA, FIPS 205) alongside their primary ML-DSA-65 key. The registration would be one-time and live in a registry trie that does not widen the account record.

The backup key would be **never used for normal transactions** — dormant unless the protocol declared an emergency algorithm rotation, at which point:

- the protocol would stop accepting signatures using the broken algorithm at a specified anchor height;
- users with a registered backup key would sign with the backup key from that anchor forward;
- the wallet UX would transition the user to the backup key transparently;
- users without a registered backup would be **frozen** — never drainable by the attacker, recoverable through a runbook-based claim process.

The frozen-account claim process would require the principal to prove control of the account through a series of out-of-band attestations (recovery contacts, KYC linkage if it existed, hardware-bound emergency tokens), documented in the recovery runbook and designed to be slow and human-checked so that the very attack that triggered the freeze could not be used to drain frozen accounts through the recovery channel. Until this mechanism returns in a future genesis, a break-day migration is instead handled by the emergency **freeze** (§23.2) plus a coordinated hard fork.

23.2 Emergency freeze — scope and limits

In the case of a coordinated systemic event — a discovered cryptographic break before rotation completes, or a confirmed adversarial fork — the chain supports an emergency **freeze** mechanism. The freeze is gated by a multi-signature Foundation key with a declared signer set and a ratification window.

What the freeze is for (the entire scope, exhaustively):

- pausing admission of new transactions during a confirmed cryptographic-primitive break, while a signature-primitive migration is coordinated as a hard fork (the in-protocol emergency-key registry this step once fed is removed — see §23.1);
- pausing transaction admission during a confirmed adversarial fork, until the canonical chain state is re-established.

Cross-chain interop is handled by an external provider outside consensus (§20), so pausing a compromised interop route is the provider's control surface, not a Monolythium freeze action.

What the freeze is not for:

- routine protocol upgrades — those happen through operator software rollout, not through the freeze mechanism;
- parameter changes — the constitutional parameters (supply cap, inflation cap, delegation cap schedule, service-weighted reward model and its per-service weights) require a coordinated hard fork, not a freeze;
- protocol-direction decisions — direction is set by maintainers and operators in public, not through the emergency mechanism;
- asset confiscation — the freeze pauses transaction admission; it does not move user funds and cannot rewrite balances;
- ongoing supervision — the freeze is time-bounded by the ratification rules and cannot be sustained as a normal operating mode;
- censorship of specific accounts — the freeze pauses the chain globally; per-account freezing is not a freeze-mechanism action.

The freeze is a circuit breaker: it exists so that the chain has a documented, accountable, time-bounded path through a worst-case event, instead of relying on improvisation under duress. It is **not** a governance backdoor. The chain's no-governance design (§4.1) and the freeze mechanism are compatible exactly because the freeze is scoped to events with clear, observable triggers (a cryptographic break or an adversarial fork) and not to ongoing decisions about how the protocol should evolve.

The signer set, signer responsibilities, ratification timeline, and post-freeze audit obligations are published and verifiable. Every freeze action is logged, justified, and re-viewable.

23.3 Multisig treasury

The Monolithium Foundation treasury is held in a multisig with a published signer set. The treasury cannot move funds without a quorum, and the quorum cannot be reduced through any in-protocol mechanism. Treasury transactions are published with rationale.

The treasury is funded from the genesis reserve (allocation category 2 in §16.2) and from buyback-burn yield over time. It does not draw on the 8% annual inflation cap.

23.4 Recovery runbooks

Recovery procedures need public, human-auditable operational playbooks before mainnet. They are not the unactivated economic runbook ABI from §18.9, and this paper does not claim that a Foundation-signed on-chain recovery template or frozen-account claim path currently exists. The removed emergency-key registry cannot be recovered through documentation; any future key-rotation mechanism requires a subsequent genesis and a separately reviewed protocol.

PART 3 — ADOPTION AND OUTLOOK

24. Developer Experience

The developer experience target is:

- Rust-first contract development;
- a Mono contract SDK;
- deterministic local testing;
- ABI and artifact generation;
- typed events and receipts;
- wallet and explorer support for MRC assets;
- contract deployment tooling;
- testing and fuzzing utilities;
- interop and market simulation tools;
- indexer-ready schemas.

The chain must replace the convenience that EVM compatibility normally provides. That means SDKs, docs, templates, wallets, explorers, and examples are not optional — they are part of the core adoption surface.

The chain is designed to be approachable for:

- Rust developers;
- AI-assisted developers using LLMs that already understand Rust well;
- fintech and payments teams;
- game and NFT builders;
- agent-platform builders;
- interop and infrastructure operators;
- teams that need post-quantum account security.

The goal is not to make EVM developers feel at home by copying Ethereum. The goal is to make a better native environment for the next category of builders.

24.1 AI runbooks for developers

The target developer experience prevents AI assistants from improvising raw financial actions. After the §18 protocol is activated, an assistant could select from released, typed, wallet-visible actions that can be simulated, approved, executed, indexed, and audited. The provisional design vocabulary includes:

- `pay_vendor`

- `open_escrow`
- `release_escrow`
- `refund_payment`
- `book_service`
- `place_trade`
- `set_spending_policy`
- `revoke_agent_permission`
- `verify_receipt`
- `rate_vendor`

These names are not current SDK methods, MCP transaction tools, or a promise that an assistant may sign. The intended bridge from natural language to settlement retains human authority in the wallet and requires the wallet and protocol to validate the same released schema.

24.2 AI-assisted contracts

The Rust ecosystem has fifteen years of training data behind it. AI coding assistants write good Rust today and will only get better. Smart-contract languages with smaller training corpora are at a structural disadvantage in the AI-assisted development era.

The chain's choice of Rust is partly a bet that the cost of writing a contract will fall fastest on the language with the deepest, cleanest training surface. The bet is not that AI will replace contract authors — it is that the marginal cost of producing a correct, well-tested, idiomatic contract will fall faster in Rust than in any alternative, and the chain wants to ride that curve.

25. User Experience

This section describes the **target** wallet and explorer experience. It is not a statement that every listed module or screen exists today. For everyday users, an eventual production chain should not feel like a research project.

The target experience lets users hold LYTH and released MRC assets, distinguish native, issuer-minted, external-provider, and private routes, and understand risk before signing. Agent spending policy, reputation, and escrow screens appear only after the corresponding §18 protocol is activated; Steele approval previews are not those economic states.

The wallet does not expose users to raw implementation details. But it does not hide important risk either.

An external interop route with weaker trust assumptions looks different from a proof-verified provider route. A wrapped asset looks different from a native issuer asset. An agent account with spending authority is easy to inspect and revoke. The bech32m address with its per-type discriminator (§13) tells the user at a glance whether they are sending to a person, a contract, a cluster, or an external-interop endpoint.

This is one of the most important adoption requirements. A non-EVM chain compensates with **clarity**.

25.1 Target four-button autovote

The v5 design proposes an **Intelligent Autovote** surface with four named modes. This is a wallet UX target, not a current shipped-feature claim; the eventual options must match the activated delegation protocol:

- **Max Yield.** Allocate to the highest-APR clusters consistent with the per-cluster cap.
- **Max Diversity.** Spread allocation across as many independent clusters as the current cap allows, weighted by reputation and uptime.
- **Max Decentralization.** Actively route stake away from clusters with high correlated-preference scores, geographic concentration, or shared operator membership.
- **Custom.** Manual per-cluster allocation, with the cap enforced at submission time and the wallet warning before any out-of-policy distribution is signed.

The four-button surface gives the user a simple choice over a complex underlying state machine. The user does not need to think about cap percentages or diversity scoring; the wallet does.

25.2 The wallet shows risk

Wallets render risk in addition to balance. Before a user signs:

- the asset's denomination (public or private);

- the asset's route (native, issuer-minted, or arrived through an external interop provider, with the provider's route and trust model);
- for assets that arrived through an external interop provider, the provider-reported route status and any provider-side risk metadata;
- the agent sub-account's policy summary if applicable;
- the runbook the assistant has selected, with parameters spelled out;
- the recipient's name (if registered) and the recipient's bech32m address;
- the depth of finality (number of confirming anchors) the receiving party expects.

Users can decline. Users can approve once, approve once with caps, or approve a long-running policy. The wallet preserves the user's authority.

26. Honest Limitations

The chain's direction is a strategic bet. The risks are real and named.

- Some developers will not migrate from Solidity to Rust.
- Liquidity may arrive more slowly without direct EVM compatibility.
- Wallets and explorers require more custom work than they would on an EVM chain.
- Native MRC standards must earn trust before they reach the ubiquity ERC standards have.
- The Rust/RISC-V contract tooling must be excellent — anything less than excellent loses to a familiar EVM environment.
- Interop depends on an external provider; the security and availability of cross-chain movement is bounded by that provider's trust model and is not verified by Monolythium consensus.
- A post-quantum application-layer zero-knowledge verifier (FRI/STARK) is still ahead: it ships gated off, and standing up on-chain-verifiable STARK receipts is real, unfinished work.
- The market for agent commerce may take longer to mature than expected.
- Post-quantum signatures are larger than classical signatures, raising storage and bandwidth costs.
- Distributed validator technology has not been deployed at the chain's target scale in a DAG-BFT configuration; the operational learning is ahead, not behind.
- The bifurcated denomination is structurally hostile to certain user expectations from existing privacy chains; users coming from those chains will find the constraints unusual.
- The composition stance with the major agent-payment standards (§3) depends on those standards remaining open and composable. A standard that closes off the integration surface would constrain the wedge for that particular rail; the chain's bet is that most of them stay open, because their open status is part of their adoption story.

These are not footnotes. They are the cost of choosing a distinct category.

The counter-risk is also real: becoming another EVM-compatible chain may make adoption easier at first but leave the project buried under stronger incumbents, deeper liquidity, and thousands of similar competitors. The chain accepts the harder path because the easier path leads to a category in which it is already too late to win.

27. What Success Looks Like

Monolythium succeeds if it becomes a credible settlement layer for:

- AI agents acting on behalf of human or organizational principals;
- human-to-agent commerce;
- agent-to-agent commerce;
- payments;
- token and NFT issuance;
- spot markets;
- cross-chain swaps;
- safer cross-chain interop through external providers;
- long-lived, post-quantum digital identities.

The early test is not whether every Solidity project deploys unchanged. That is not the chosen market.

The early test is whether users and builders understand the value of safer cross-chain interop, visible route risk, native agent permissions, Rust-native contracts, post-quantum accounts, clean token and market standards, and a chain designed around the next decade rather than the previous one.

Success is also measurable along structural dimensions that are independent of any single market hype cycle:

- the number of independent clusters and the geographic and ASN distribution of their operators;
- the depth of the discovery registry and the number of legitimate providers it hosts;
- the volume of escrowed agent-to-agent and human-to-agent transactions;
- the share of cross-chain volume moving through proof-verified external-provider routes versus trusted-multisig routes;
- the volume of agent-payment-standard composition (x402, AP2, ACP, MCP) settling against Monolythium spending policies and escrows;
- the diversity of MRC asset issuers;
- the survival of agent identities across model providers — that is, the structural portability of reputation.

A network that scores well on these is a network that has delivered what the design promised. A network that scores well on token price alone has not.

28. Closing

Monolythium is a deliberate break from the default Layer-1 playbook.

It does not try to win by becoming a slightly faster EVM chain. It chooses Rust on a deterministic RISC-V target, post-quantum accounts as default, native asset standards, native markets, a target agent-commerce architecture, external-provider interop, a lean post-quantum core, a structurally separated privacy design, a public cluster marketplace, and a smaller protocol surface.

Its target architecture composes underneath major agent-payment standards rather than fighting them, aiming to provide a chain-anchored policy, escrow, identity, and reputation layer after each capability is activated.

That choice is harder. It means more tooling, more education, and a slower path to existing Solidity liquidity. It also gives the chain a clearer identity and a stronger foundation for the workloads it expects to serve.

The thesis is straightforward: the next major settlement layer will not be the chain that copies Ethereum most closely. It will be the chain that gives new economic actors — autonomous agents, organizations delegating to software, services that want neutral cross-platform rails — safer rails to transact across applications, markets, and jurisdictions.

Monolythium is being built toward that future. The dated capability and network-status boundaries in this reconciliation define what can be relied on today.

Acknowledgments

The Monolythium protocol is the product of years of design work, research, and engineering by the team at Mono Labs R&D LLC, in coordination with the Monolythium Foundation. The protocol draws on a wide body of academic and open-source work, including:

- the Dilithium and Kyber post-quantum primitives standardized by NIST as ML-DSA and ML-KEM;
- the Starfish family of leaderless DAG-BFT consensus protocols;
- the BIP-39 wordlist and bech32m encoding conventions;
- the FRI-based proof-system family and the broader zero-knowledge research community;
- the BLAKE3 hash function;
- the open agent-payment standards that the target architecture may compose against after compatible integrations are released — x402, AP2, ACP, and MCP — whose existence makes the settlement-layer wedge concrete and addressable.

This whitepaper is a synthesis of those contributions into a specific Layer-1 architecture. Errors are the authors'. Attribution should be given to the project, not to the underlying primitives whose authors deserve their own citation.

Entities

Monolythium is operated and stewarded by two entities:

- **Mono Labs R&D LLC** — the operating company, based in San Francisco, California. Mono Labs R&D LLC develops the Monolythium protocol, operates the reference client and SDK, and is the seller of record for the LYTH token in jurisdictions where a seller of record is required.
- **Monolythium Foundation** — the protocol steward, based in the Cayman Islands. The Foundation operates the Foundation multisig treasury, the genesis name reserve, the emergency-freeze ratification window, and other constitutional-layer functions that require an entity rather than a company. (The emergency-key registry it would have stewarded is removed — see §23.1.)

The two entities are independent, with separate governance, separate financial accounts, and separate roles in the protocol's operation.

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